AN INTERNSHIP REPORT ON

"COUNTRIES RANKING ANALYSIS USING POWER BI"

Submitted in accordance with the requirement for the degree of

"BACHELOR OF SCIENCE"

Submitted by

VAMSIDHAR MANGENA

K2000428

Under the Guidance of

M. Kala Devi, M.C.A, M. Tech (C.S.E)

Department Of Computer Science

KAKARAPARTI BHAVANARAYANA COLLEGE(Autonomous)

(Sponsored by S.K.P.V.V.Hindu High School)

Kothapet, Vijayawada

PROGRAM BOOK FOR

VI SEMESTER INTERNSHIP

Name of the Student: ***VAMSIDHAR MANGENA***

Name of the College: ***KAKARAPARTI BHAVANARAYANA COLLEGE***
(Autonomous)

Registration Number: ***K2000428***

Period of Internship: From ***09/01/2023*** To ***10/05/2023***

Name &Address of the Intern Organization:

Quality Thoughts Infosystems – Vijayawada
Door no 3-71, Sri Kurapati Enclave C/o Kurapati Ravuri Satyanarayana St,
opposite Tamilnadu Mercantile Bank, Gollapudi, Vijayawada - 521225

Krishna University

YEAR

2023

An Internship Report on

COUNTRIES RANKING ANALYSIS REPORT

Submitted in accordance with the requirement for the degree of

Bachelor Of Science

Under the Faculty Guideship of

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Department of Computer Science

Kakaraparti Bhavanarayana College(A)

Submitted by:

VAMSIDHAR MANGENA

Reg. No: K2000428

Department of Computer Science

Kakaraparti Bhavanarayana College(A)

Student's Declaration

I, ***VAMSIDHAR MANGENA*** a student of BSC (MSCs) Program, Reg No. *K2000428* of the Department of Computer Science College do hereby declare that I have completed the mandatory internship from 9th January 2023 to 10th May 2023 in **Quality Thought** under the Faculty Guideship of M. Kala Devi MCA, M. Tech, Department of Computer Science, Kakaraparti Bhavanarayana College.

(Signature and Date)

Official Certification

This is to certify that **M.Vamsidhar** Reg no. *K2000428* has completed his internship in ***Quality Thought*** *on* ***COUNTRIES RANKING ANALYSIS USING POWER BI*** under my supervision as a part of the partial fulfillment of the requirement for the Degree of BSC (MSCs) in the Department of Computer Science,

karaparti Bhavanaraya College.

(Signatory with Date and Seal)

Endorsements

Faculty Guide

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that M. ***VAMSIDHAR*** Reg No. *K2000428* of Kakaraparti Bhavanarayana College underwent a n internship in *Quality thought* from 09th January 2023 to 10th May 2023. The overall performance of the intern during his/her internship is found to be ____________________ Satisfactory/not Satisfactory.

Authorized Signatory with Date and Seal

Acknowledgments

I take this opportunity to express our profound gratitude and deep regards to our guide **M. KALA DEVI, M.C.A, M. Tech** for this exemplary monitoring project, guiding and correcting various documents of mine with attention and care. The help and guidance given by her from time to time shall carry us a long way in the journey of life on which we are about to embark.

I would also like to express my profound gratitude to our project guides from **QUALITY THOUGHT** for helping us to do the project work. They have taken pain to go through the project and made necessary corrections as and when needed, and constant encouragement throughout the course of this project report a would like to express my heartfelt gratitude to our obliged staff member **P. RAVINDRA, M.C.A, M. Tech, HOD** of Computer Science for their valuable help, suggestion, and feedback.

Contents

LEARNING OBJECTIVES

On completing this course, as learners, we will be able to:

- Identify the primary components of the Power BI interface: reports, data, and model views
- Import Excel data and build basic visuals
- Publish a desktop report to the Power BI Service
- Identify common challenges in Power BI data models, implement smart solutions, and avoid common mistakes

LEARNING OUTCOMES

This Power BI training course will help you get the most out of Microsoft's Power BI, a suite of tools that lets you build interactive dashboards for analyzing data and extracting business insights. This course will help you master the development of dashboards from published reports, discover greater insights from your data with Quick Insights, and learn practical applications for Power BI tasks, such as gathering and analyzing data. You will also learn valuable Power BI troubleshooting tips.

- Understand Power BI concepts like Microsoft Power BI desktop layouts, BI reports, dashboards, and Power BI DAX commands and functions
- Gain a competitive edge in creating customized visuals and deliver a reliable analysis of vast amounts of data using Power BI
- Learn how to experiment, fix, prepare, and present data quickly and easily
- Create a sales analysis report and a project management report
- Form relationships in your data model and learn data visualization best practices

Quality Management Training, Corporate Training Providers

Quality Thought Technologies provides ultimate software training. Our group is technical savvier who are dedicated bringia nging difference in the way people are trained in software tools & technologies. We focus on practical approach with well-defined process & framework that perfectly transforms the academics to professionals. We believe in Practical Approach than Theoretical Approach.

In Quality Thought we change each penny of yours to figure up. Accompanied by a motive to furnish extreme branch of subject knowledge to every individual trainee of ours, Quality Thought, a pioneer in the IT training sector higher up than a decade experienced in training and placements. Our team follows excessive dedication and motivation, we persuade quality of excellence and accomplished fulfilment in our training programs. Quality Thought is the gateway to a blazing future in IT sector. We market leaders in IT training space and are pioneer's in introducing brand new courses, products and services. Accompanied specialization in IOT and Data Science, numerous programs in future.

We are always maintaining our ascendance by being the first to introduce brand new products and services in the industry. We are mainly specializing in career training in the area off software quality assurance. We make the trainee dream into reality. Quality Thought educational and professional development organization that has been founded to advance the software test and quality assurance profession by promoting and recognizing professionalism through education, consulting Services

At Quality Thought you get:

- Amplify your profession in Quality Thought.
- Experience Training of Excellence from our bureaucrats.
- Modernize persistently in Learning New Skills which presents You Beneficial Utility.
- Accessibility of a unique training in magnificent ambience.
- Obtainability of our faculty members for perpetual timing for assistance.
- Premier direction and prop up by our HR-Team members in placing our trainees in foremost organizations.

Quality Mission Statement

To furnish a training quality of excellence, career orientation and to attain the goal of bagging opportunities in IT sector. Quality Thought fulfills its Mission through a disciplined approach. It provides only real time services to its customers though its real time faculty. As Part of Its Mission and Vision Quality Thought:

Quality Thought vision is to bring new insights in your individual and organizational search & transform your career. Our mission is to provide the best services to all with value addition and cutting edge in your job search, thereby saving your precious time.

- It Promotes and Provides education of software testing professionals around the world.
- It creates a pool of qualified software testing professionals to meet the needs of testing organizations.
- It Provides assistance and guidance to members, both corporate and individuals, performing testing of all types of software systems.

- It provides a framework for assessing organizational testing practices and procedures.
- It Provides services, methods, and tools supporting the discipline of software testing.
- Provide an open forum for discussing different testing issues.

Quality Vision Statement

"Quality Thought, persistently focuses on spotting advanced and emerging technologies and assembles programs for forthcoming requisites".

Quality Thought vision is to bring new insights in your individual and organizational search & transform your career. Our mission is to provide the best services to all with value addition and cutting edge in your job search, thereby saving your precious time. It Promotes and Provides education of software testing professionals around the world. It creates a pool of qualified software testing professionals to meet the needs of testing organizations. It Provides assistance and guidance to members, both corporate and individuals, performing testing of all types of software systems. It provides a framework for assessing organizational testing practices and procedures. It Provides services, methods, and tools supporting the discipline of software testing. Provide an open forum for discussing different testing issues.

Quality Policy

Quality Thought educational and professional development organization that has been founded to advance the software test and quality assurance profession by promoting and recognizing professionalism through education, consulting Services. Quality Thought fulfills its Mission through a disciplined approach. It provides only real time services to its customers though its real time faculty.

Quality Thought provides the best in industry classroom driven training programs with live trainings conducted by subject matter experts. At QT we have excellent class room infrastructure for a conducive face-to-face training programs to be conducted. Our classroom programs are well planned with optimal batch sizes and individual attention on every participant in a well-mannered environment ensuring a healthy learning platform for all our students. Fresh graduates as well as working professional can enroll our multitude of classroom training programs and achieve their professional career goals.

Activities/Responsibilities in the Intern Organization during Internship

Microsoft Power BI is a collection of applications, software services and connectors that work together to turn your unrelated sources of data into clear, visually immersive, and interactive insights. It is a business analytics interactive service provided by Microsoft that helps you to connect easily with your data sources, and visualize.

With the help of Power BI, individuals can get data from a wide range of systems in the cloud and create dashboards that will track the metrics they care the most. With this technology, enterprises can see their business performance more closely and get immediate results with real-time dashboards available for every device. By taking our Microsoft Power BI training course, anyone can become a competent data analyst. All the topics and units are broken down in an easy to learn manner, making the course extremely enjoyable and our friendly instructors delight all our learners to use Power BI successfully. The user-friendly Power BI reporting application has been designed to help users, of all experiential levels, produce insightful data analysis - hence our interactive and highly engaging BI training course helps candidates across departments communicate performance-related data in a visually understandable manner.

Power BI is a business analytics solution owned by Microsoft. Analysts use it for data visualization, draw insights from it, and share the dashboards and visual reports across the organization. Moreover, one can access these reports sitting at their workplace or anywhere through Power BI apps offered by Microsoft.

Getting Data

Getting Data is the first experience of **working with Power BI**. You can connect to many data sources on-premises or on cloud. For some data sources you can have a live or direct connection, for some connection works offline. For some connections you need a gateway or connector to be installed.

Power Query for Data Transformation

Data analysis and BI (Business Intelligence) world starts from data extraction and transformation. Power Query is the data transformation engine of Microsoft Power BI. Power Query comes as part of Excel 2016, or as an add-in for Excel 2013 and 2010.

Power Query is used to transform the data to consolidate it like Merging, Appending, Creating Custom Columns, Creating Conditional Columns, Applying Aggregations, sorting if required and removing unwanted data. To work on Power Query, make sure, you should be connected to SOURCE else you can get data into Power Query.

Data Modeling and Data Analysis Expression (DAX)

Power Pivot is x Velocity in-memory data modelling engine of the Power BI. Modelling effectively is the key of high-performance BI solution. Data Modeling is used to create relationships, and calculated members, as well as advanced best practices and **DAX expressions**. DAX is Data Analytical Expression language. DAX has similar structure to excel functions, but it is different. You will learn basic functions as well as complex functions and scenarios of using them in real world challenges.

DAX or **Data Analysis Expression** is the language for querying and creating calculated measure and columns and tables in Power BI

At the end of this training, we will be able to design the proper data model in Power BI, understand all relationship requirements and implement the right relationship, write complex DAX expressions for your analytics need, and put them all together to build the best model for your data analysis solution using Power BI.

Data Visualization - Reports

Data Visualization is the front end of any BI application, this is the user viewpoint of your system. It is critical to visualize measures, and dimensions effectively so the BI system could tell the story of the data clearly. We will be trained conceptual best practices of data visualizations which is valid through all data visualization tools. We will be trained Power View and Power Map skills. Power View is the interactive data visualization tool and Power Map is 3D geo-spatial data visualization tool. We will be learning how to create effective charts, and dashboards using these tools as well as best practices for working with these tools.

ACTIVITY LOG

ACTIVITY LOG FOR THE FIRST WEEK

Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
09/01/2023 MONDAY	Introduction class- description of what is learnt through the internship.	Importance of business intelligence tools.	
10/01/2023 TUESDAY	Introduction to operating systems – LAN, WAN, MAN, PAN	Basic knowledge of operating systems.	
11/01/2023 WEDNESDAY	Working of an Operating System – Firewall Layers – VPN – IP address of a system	Internet protocols (IPs) – How security of a computer works.	
12/01/2023 THURSDAY	Installation of POWER BI desktop application - SAAS, IAAS, PAAS	Software a s a service in power BI.	
13/01/2023 FRIDAY	BHOGI HOLIDAY	------	
14/01/2023 SATURDAY	SANKRANTHI HOLIDAY	------	

WEEKLY REPORT

WEEK – 1 (From Dt 09-01-2023 to Dt 14-01-2023)

An operating system is a program that acts as an interface between the computer user and computer hardware, and controls the execution of programs. The operating system (OS) manages all of the software and hardware on the computer. It performs basic tasks such as file, memory and process management, handling input and output, and controlling peripheral devices such as disk drives and printers.

Most of the time, there are several different computer programs running at the same time, and they all need to access your computer's central processing unit (CPU), memory and storage. The operating system coordinates all of this to make sure each program gets what it needs.

A **Computer Network** is a group of two or more interconnected computer systems that use common connection protocols for sharing various resources and files. You can establish a computer network connection using either cable or wireless media. Every network involves hardware and software that connects computers and tools.

There are various types of Computer Networking options available. The classification of network in computers can be done according to their size as well as their purpose.

The size of a network should be expressed by the geographic area and number of computers, which are a part of their networks. It includes devices housed in a single room to millions of devices spread across the world. Following are the popular types of Computer Network:

- **PAN** (Personal Area Network) is a computer network formed around a person. It generally consists of a computer, mobile, or personal digital assistant. PAN can be used for establishing communication among these personal devices for connecting to a digital network and the internet.
- A **Local Area Network** (LAN) is a group of computer and peripheral devices which are connected in a limited area such as school, laboratory, home, and office building. It is a widely useful network for sharing resources like files, printers, games, and other application. The simplest type of LAN network is to connect computers and a printer in someone's home or office. In general, LAN will be used as one type of transmission medium. It is a network which consists of less than 5000 interconnected devices across several buildings.
- **WAN** (Wide Area Network) is another important computer network that which is spread across a large geographical area. WAN network system could be a connection of a LAN which connects with other LAN's using telephone lines and radio waves. It is mostly limited to an enterprise or an organization.
- A **Metropolitan Area Network** or MAN is consisting of a computer network across an entire city, college campus, or a small region. This type of network is large than a LAN, which is mostly limited to a single building or site. Depending upon the type of configuration, this type of network allows you to cover an area from several miles to tens of miles.

VPN stands for "Virtual Private Network" and describes the opportunity to establish a protected network connection when using public networks. VPNs encrypt your internet traffic and disguise your online identity. This makes it more difficult for third parties to track your activities online and steal data.

An **Internet Protocol (IP)** address is a unique numerical identifier for every device or network that connects to the internet. Typically assigned by an internet service provider (ISP), an IP address is an online device address used for communicating across the internet.

An IP address serves two principal functions: it identifies the host, or more specifically its network interface, and it provides the location of the host in the network, and thus the capability of establishing a path to that host. Its role has been characterized as follows: "A name indicates what we seek. An address indicates where it is. A route indicates how to get there."[2] The header of each IP packet contains the IP address of the sending host and that of the destination host.

IaaS, PaaS and SaaS are the three most popular types of cloud service offerings. They are sometimes referred to as cloud service models or cloud computing service models.

- IaaS, or infrastructure as a service, is on-demand access to cloud-hosted physical and virtual servers, storage and networking - the backend IT infrastructure for running applications and workloads in the cloud.

- PaaS, or platform as a service, is on-demand access to a complete, ready-to-use, cloud-hosted platform for developing, running, maintaining and managing applications.

- SaaS, or software as a service, is on-demand access to ready-to-use, cloud-hosted application software.

IaaS, PaaS and SaaS are not mutually exclusive. Many mid-sized businesses use more than one, and most large enterprises use all three.

'As a service' refers to the way IT assets are consumed in these offerings - and to the essential difference between cloud computing and traditional IT. In traditional IT, an organization consumes IT assets - hardware, system software, development tools, applications - by purchasing them, installing them, managing them and maintaining them in its own on-premises data center. In cloud computing, the cloud service provider owns, manages and maintains the assets; the customer consumes them via an Internet connection, and pays for them on a subscription or pay-as-you-go basis.

So, the chief advantage of IaaS, PaaS, SaaS or any 'as a service' solution is economic: A customer can access and scale the IT capabilities it needs for a predictable cost, without the expense and overhead of purchasing and maintaining everything in its own data center. But there are additional advantages specific to each of these solutions.

ACTIVITY LOG FOR THE SECOND WEEK

Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
16/01/2023 MONDAY	GET DATA – from excel to Power BI. Extracting Data	Different sources of getting data to power bi desktop.	
17/01/2023 TUESDAY	Introduction to visualization charts - Formatting of Visuals – Stacked bar chart.	Uses of stacked bar chart – applications - where to use it.	
18/01/2023 WEDNESDAY	Stacked column chart – formatting of a stacked column chart – Power BI - Excel Differences	Use of stacked column chart – how to format it – its uses	
19/01/2023 THURSDAY	Cluster – Bar chart/column chart – load balance – applications to use a cluster chart.	Uses of cluster column and bar chart – formatting it	
20/01/2023 FRIDAY	Seminar by students	Recap of what has been learnt so far	
21/01/2023 SATURDAY	Line chart and its formatting settings	Uses of line chart – its advantages – formatting a line chart.	

WEEKLY REPORT

WEEK – 2 (From Dt 16-01-2023 to Dt 21-01-2023)

Power BI desktop app is used to create reports, while Power BI Services (Software as a Service - SaaS) is used to publish the reports, and Power BI mobile app is used to view the reports and dashboards.

Minimum requirements

The following list provides the minimum requirements to run Power BI Desktop:

Important

Power BI Desktop is no longer supported on Windows 7.

- Windows 8.1 or Windows Server 2012 R2 or later.
- .NET 4.6.2 or later.
- Microsoft Edge browser (Internet Explorer is no longer supported)
- Memory (RAM): At least 2 GB available, 4 GB or more recommended.
- Display: At least 1440x900 or 1600x900 (16:9) required. Lower resolutions such as 1024x768 or 1280x800 aren't supported because some controls (such as closing the startup screens) display beyond those resolutions.
- Windows display settings: If you set your display to change the size of text, apps, and other items to more than 100%, you won't see some dialogs that you must interact with to continue using Power BI Desktop. If you encounter this issue, check your display settings in Windows by going to **Settings** > **System** > **Display**, and use the slider to return display settings to 100%.
- CPU: 1 gigahertz (GHz) 64-bit (x64) processor or better recommended.
- WebView2: If WebView2 wasn't automatically installed with Power BI Desktop or if it was uninstalled, download and run the installer for WebView2

Install as an app from the Microsoft Store

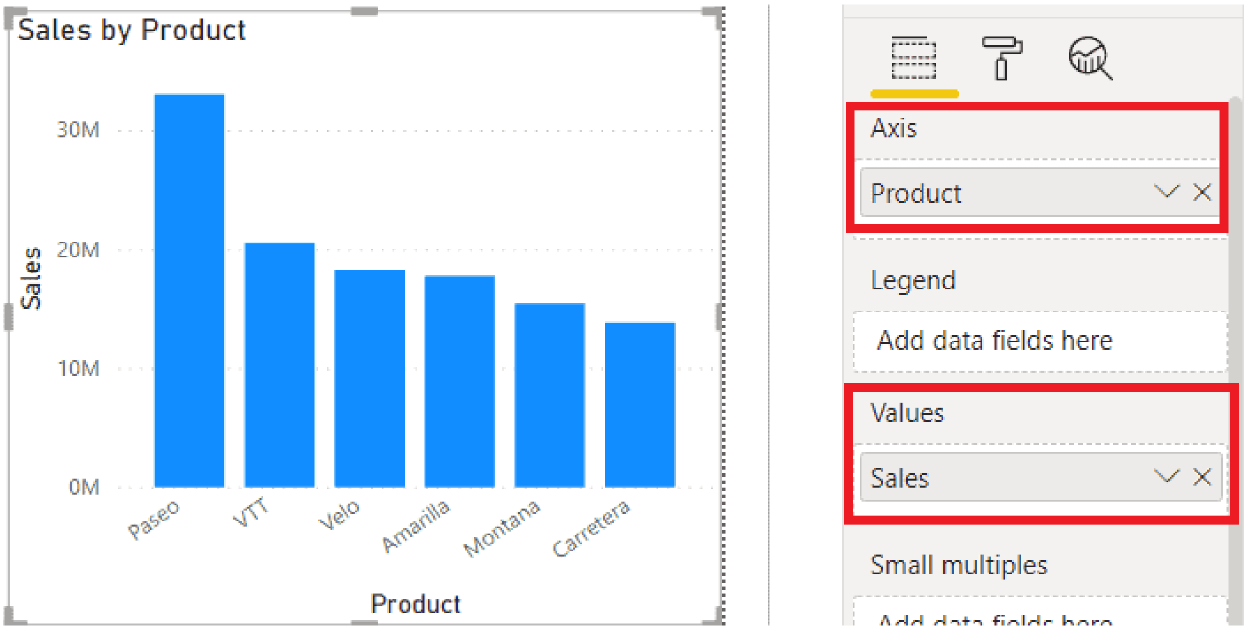
- Open a browser and go directly to the Power BI Desktop page of the Microsoft Store.
- From the Power BI service, in the upper right corner, select the Download icon and then choose Power BI Desktop.
- Go to the Power BI Desktop product page, and then select Download Free.

Get data - The Power BI Desktop makes discovering data easy. You can import data from a wide variety of data sources. After you connect to a data source, you can shape the data to match your analysis and reporting needs.

Power BI Stacked Column chart

A **Power BI Stacked Column chart** based on **column bars**, which comprise one or multiple legends. In a Stacked Column chart, data series are stacked one on top of the other in vertical columns. Stacked column charts show changes over time and also it is easy to compare total column length.

In a Stacked Column chart, the **Axis is represented on X-axis**, and **Data is represented on Y-axis**. Let's have a look at the below example:



In the above example, we can see the **Axis(Product)** represented on **X-axis**, whether **Y-axis** representing the **data/values(Sales)**.

When to use Stacked Column chart in Power Bi

Now, we will discuss in which cases we can use this Stacked Column chart to visualize the data. The cases are given below:

- The Stacked Column charts are great to represent the parts of multiple totals.
- This chart work well for only a few totals.
- The Stacked Column charts can work great for showing data over Date.
- We can use this chart for representing time data, having same intervals.
- This chart works better if the total have short labels.

Power BI Stacked Bar Chart

Power BI Stacked Bar Chart Bar Chart is useful for comparing multiple dimensions against a single measure. Let me show you how to Create a Stacked Bar Chart in Power BI with examples.

How do you make a stacked bar chart in Power BI?

Click on the Stacked Bar icon under the Visualization will automatically create a dummy chart. Furthermore, adding a Numeric measure (sales) to X-axis, a dimension to Y-axis, and another dimension (stacked) to the legend section generates the actual stacked bar chart.

Bar charts are best used when showing comparisons between categories. Typically, the bars are proportional to the values they represent and can be plotted either horizontally or vertically. One axis of the chart shows the specific categories being compared, and the other axis represents discrete values.

A **clustered bar chart** is a horizontal chart, which could present multiple bars in the form of a cluster. The horizontal bars basically group together, as they are under the same, y values. We have various options to format clustered bar charts, we can change the value of the x-axis, y-axis, its title, etc. In this article, we will learn how to format a clustered bar chart in Power BI and explore its various options.

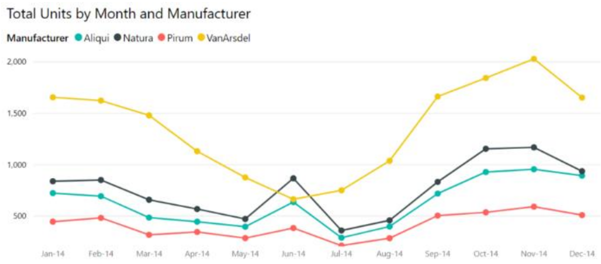
ACTIVITY LOG FOR THE THIRD WEEK

Day & Date	**Brief description of the dailyactivity**	**Learning Outcome**	**Person In-Charge Signature**
23/01/2023 MONDAY	Area chart – formatting an area chart – Row level security – DAX query	Uses of area chart – applications of area chart.	
24/01/2023 TUESDAY	Inserting a text box - line chart – stacked column chart - Seminar	Applying charts to data - Testing different charts to data	
25/01/2023 WEDNESDAY	Line chart – Clustered column chart – ribbon chart – waterfall chart – funnel chart	Uses of all the charts that have been explained.	
26/01/2023 THURSDAY	REPUBLIC DAY	------	
27/01/2023 FRIDAY	Funnel chart review – uses – advantages – where not to use a funnel chart	Using different charts for visualization – how to use different attributes of a visual - formatting	
28/01/2023 SATURDAY	VLOOKUP in excel – Task given – Seminar	Worked on what has been given as a task	

WEEKLY REPORT

WEEK – 3 (From Dt 23-01-2023to Dt 28-01-2023)

A line chart is a series of data points that are represented by dots and connected by straight lines. A line chart may have one or many lines. Line charts have an x and a y axis.



The basic area chart (also known as layered area chart) is based on the line chart. The area between axis and line is filled with colors to indicate volume.

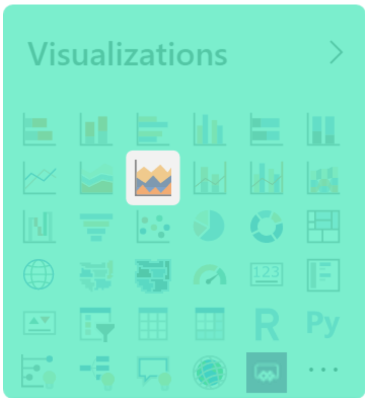
Area charts emphasize the magnitude of change over time, and can be used to draw attention to the total value across a trend. For example, data that represents profit over time can be plotted in an area chart to emphasize the total profit.

When to use a basic area chart

Basic area charts are a great choice:

- to see and compare the volume trend across a time series.
- for an individual series representing a physically countable set.

Power BI Stacked Area Chart



The **Stacked Area chart** is an extension to the **Area Chart**. As with the Area Chart, the Stacked Area chart is often used to visualise a trend in data over intervals of time, the difference is that the Stacked Areas are used to display how several quantities combine to make a whole, as it shows how each part or category contributes to the whole amount. Stacked Area Chart is plotted in the form of several area series stacked on top of one another. The height of each series is determined by the value in each data point. A typical use case for Stacked Area Charts is analyzing how each of several variables and their totals vary, on the same graphic.

In the case that we want to compare the values between groups, we end up with an overlapping area chart. In an overlapping area chart, we start with a standard line chart. For each group, one point is plotted at each horizontal value with height indicating the group's value on the vertical axis variable; a line connects all of a group's points from left to right. The area chart adds shading between each line to a zero baseline. Since the shading for groups will usually overlap to some extent, some transparency is included in the shading so that all groups' lines can be easily seen. The shading helps to emphasize which group has the largest value based on which group's pure color is visible.

Be careful that one series is not always higher than the other, or the plot may become confused with the other type of area chart: the stacked area chart. In those cases, just keeping to the standard line chart will be a better choice.

Generally, when the term 'area chart' is used, what is actually implied is the stacked area chart. In the overlapping area chart, each line was shaded from its vertical value to a common baseline. In the stacked area chart, lines are plotted one at a time, with the height of the most recently-plotted group serving as a moving baseline. As such, the fully-stacked height of the topmost line will correspond to the total when summing across all groups.

You will use a stacked area chart when you want to track not only the total value, but also want to understand the breakdown of that total by groups. Comparing the heights of each segment of the curve allows us to get a general idea of how each subgroup compares to the other in their contributions to the total.

A **funnel chart** helps you visualize a linear process that has sequential, connected stages. For example, a sales funnel that tracks customers through stages: Lead > Qualified Lead > Prospect > Contract > Close. At a glance, the shape of the funnel conveys the health of the process you're tracking.Each funnel stage represents a percentage of the total. So, in most cases, a funnel chart is shaped like a funnel -- with the first stage being the largest, and each subsequent stage smaller than its predecessor. A pear-shaped funnel is also useful -- it can identify a problem in the process. But typically, the first stage, the "intake" stage, is the largest.

You can create **ribbon charts** to visualize data, and quickly discover which data category has the highest rank (largest value). Ribbon charts are effective at showing rank change, with the highest range (value) always displayed on top for each time period.Ribbon charts connect a category of data over the visualized time continuum using ribbons, enabling you to see how a given category ranks throughout the span of the chart's x-axis (usually the timeline).

A **waterfall chart** is a form of data visualization that helps in understanding the cumulative effect of sequentially introduced positive or negative values. These intermediate values can either be time based or category based. The waterfall chart is also known as a **flying bricks chart** or **Mario chart** due to the apparent suspension of columns (bricks) in mid-air. Often in finance, it will be referred to as a **bridge**.Complexity can be added to waterfall charts with multiple total columns and values that cross the axis. Increments and decrements that are sufficiently extreme can cause the cumulative total to fall above and below the axis at various points. Intermediate subtotals, depicted with whole columns, can be added to the graph between floating columns.

ACTIVITY LOG FOR THE FORTH WEEK

Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
30/01/2023 MONDAY	Column profiling – column quality – column distribution – description – datatype of columns	Learnt about the components of a column	
31/01/2023 TUESDAY	Creating a parameter – pie chart – formatting – transforming data.	Transforming data - modelling of data – uses of pie chart.	
01/02/2023 WEDNESDAY	Introduction to field parameters – Applying parameters to visuals.	Uses of fields parameters – applications of fields parameters	
02/02/2023 THURSDAY	Seminar by students	Recap of what we learnt so far	
03/02/2023 FRIDAY	Back to parameters and field parameters – trying them out with different data	Creating – working – understanding parameters.	
04/02/2023 SATURDAY	Different types of parameters – how to use them - advantages	Learnt how to use field parameter – normal parameter uses	

WEEKLY REPORT

WEEK – 4 (From Dt 30-01-2023 to Dt 04-02-2023)

Using the data profiling tools

The data profiling tools provide new and intuitive ways to clean, transform, and understand data in Power Query Editor. They include:

- Column quality
- Column distribution
- Column profile

To enable the data profiling tools, go to the **View** tab on the ribbon. Enable the options you want in the **Data preview** group, as shown in the following image.

After you enable the options, you'll see something like the following image in Power Query Editor.

By default, Power Query will perform this data profiling over the first 1,000 rows of your data. To have it operate over the entire dataset, check the lower-left corner of your editor window to change how column profiling is performed.

Column quality

The column quality feature labels values in rows in five categories:

- **Valid**, shown in green.
- **Error**, shown in red.
- **Empty**, shown in dark grey.
- **Unknown**, shown in dashed green. Indicates when there are errors in a column, the quality of the remaining data is unknown.
- **Unexpected error**, shown in dashed red.

These indicators are displayed directly underneath the name of the column as part of a small bar chart, as shown in the following image.

The number of records in each column quality category is also displayed as a percentage.By hovering over any of the columns, you are presented with the numerical distribution of the quality of values throughout the column. Additionally, selecting the ellipsis button (...) opens some quick action buttons for operations on the values.

Column distribution

This feature provides a set of visuals underneath the names of the columns that showcase the frequency and distribution of the values in each of the columns. The data in these visualizations is sorted in descending order from the value with the highest frequency.By hovering over the distribution data in any of the columns, you get information about the overall data in the column (with distinct count and unique values). You can also select the ellipsis button and choose from a menu of available operations.

Column profile

This feature provides a more in-depth look at the data in a column. Apart from the column distribution chart, it contains a column statistics chart. This information is displayed underneath the data preview section, as shown in the following image.

A parameter serves as a way to easily store and manage a value that can be reused.

Parameters give you the flexibility to dynamically change the output of your queries depending on their value, and can be used for:

- Changing the argument values for particular transforms and data source functions.
- Inputs in custom functions.

You can easily manage your parameters inside the **Manage Parameters** window. To get to the **Manage Parameters** window, select the **Manage Parameters** option inside **Manage Parameters** in the **Home** tab.

Parameter properties

A parameter stores a value that can be used for transformations in Power Query. Apart from the name of the parameter and the value that it stores, it also has other properties that provide metadata to it. The properties of a parameter are:

- **Name**: Provide a name for this parameter that lets you easily recognize and differentiate it from other parameters you might create.
- **Description**: The description is displayed next to the parameter name when parameter information is displayed, helping users who are specifying the parameter value to understand its purpose and its semantics.
- **Required**: The checkbox indicates whether subsequent users can specify whether a value for the parameter must be provided.
- **Type**: Specifies the data type of the parameter. We recommended that you always set up the data type of your parameter. To learn more about the importance of data types, go to Data types.
- **Suggested Values**: Provides the user with suggestions to select a value for the **Current Value** from the available options:
- **Any value**: The current value can be any manually entered value.
- **List of values**: Provides you with a simple table-like experience so you can define a list of suggested values that you can later select from for the **Current Value**. When this option is selected, a new option called **Default Value** will be made available. From here, you can select what should be the default value for this parameter, which is the default value shown to the user when referencing the parameter. This value isn't the same as the **Current Value**, which is the value that's stored inside the parameter and can be passed as an argument in transformations. Using the **List of values** provides a drop-down menu that's displayed in the **Default Value** and **Current Value** fields, where you can pick one of the values from the suggested list of values.

ACTIVITY LOG FOR THE FIFTH WEEK

Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
06/02/2023 MONDAY	Power query editor – making changes to the data – transform data – changing datatype.	Various ways of transforming data through power query editor	
07/02/2023 TUESDAY	What if parameters – Numeric range of parameters – table visual – DAX query for installment calculation.	Writing a DAX query – interest calculation as a query.	
08/02/2023 WEDNESDAY	Seminar by students	Recap of various concepts that were learnt.	
09/02/2023 THURSDAY	Numeric range parameters – date range parameters - applications of the respective.	Various cases for the use of date range and numeric range parameters.	
10/02/2023 FRIDAY	Seminar by students	Recap of what has been learnt so far	
11/02/2023 SATURDAY	Differences between the parameters – how to create a what if parameter - uses	Applying normal parameter – field parameter – what if parameters	

WEEKLY REPORT

WEEK – 5 (From Dt 06-02-2023 to Dt 11-02-2023)

Here's how Power Query Editor appears after a data connection is established:

- In the ribbon, many buttons are now active to interact with the data in the query.
- In the left pane, queries are listed and available for selection, viewing, and shaping.
- In the center pane, data from the selected query is displayed and available for shaping.
- The **Query Settings** pane appears, listing the query's properties and applied steps.

The ribbon in Power Query Editor consists of four tabs: **Home**, **Transform**, **Add Column**, **View**, **Tools**, and **Help**. The **Home** tab contains the common query tasks. The **Transform** tab provides access to common data transformation tasks, such as:

- Adding or removing columns
- Changing data types
- Splitting columns
- Other data-driven tasks

The **Add Column** tab provides more tasks associated with adding a column, formatting column data, and adding custom columns. The following image shows the **Add Column** tab. The **View** tab on the ribbon is used to toggle whether certain panes or windows are displayed. It's also used to display the Advanced Editor. The following image shows the **View** tab. It's useful to know that many of the tasks available from the ribbon are also available by right-clicking a column, or other data, in the center pane.

Using **What If parameter** in Power Bi can easily give you the ability to dynamically transform your data. For example, using this parameter will allow to demonstrates how your data change under various scenarios. For example how much revenue would you have if your products were at 10%, 20% or 30% of the retail price. You could add additional logic to show how this changes demand. Another scenario would be to show create a marketing mix to show how profit would change due to different investment in each channel.

How to use What IF parameters in Power BI.

- Click the Modeling tab in the top ribbon.
- Click the What IF parameter from the top ribbon.
- The What If parameter window will open
- Provide details such as Name, Data Type, Minimum, Maximum and Default number.
- Lastly, you can add an optional slicer.
- A table with a calculated measure will be created.

The Numeric range Parameter is a feature that allows us to interact with the variable as a slicer and visualize and quantify different key values in the reports. We can very easily build a Power BI Report by creating the Numeric range Parameter.

ACTIVITY LOG FOR THE SIXTH WEEK

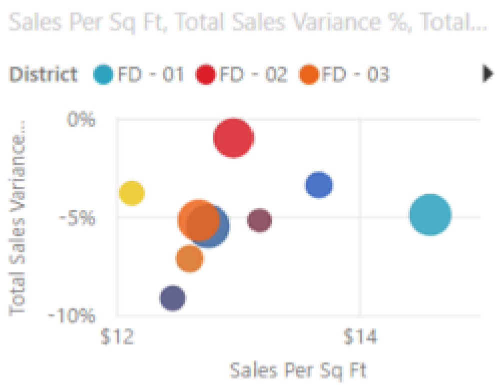
Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
13/02/2023 MONDAY	Scatter chart – functionalities – Tree graph – functionalities.	Uses of scatter chart and tree graph – applications of those charts	
14/02/2023 TUESDAY	Communication and soft skills – Ability – Written test	testing of skills – written communication.	
15/02/2023 WEDNESDAY	Soft skills introduction class – learning - types of communication	Interactive class – explanation through modules – skill building	
16/02/2023 THURSDAY	Communication and soft skills day – 2 – learning etiquettes – formal attire	How to communicate – importance of non-verbal communication	
17/02/2023 FRIDAY	Interactive class with HR – basic knowledge of work space environment – common interview questions – technical questions	Gained knowledge on questions asked during interview and how to answer them.	
18/02/2023 SATURDAY	Formatting a map visual – map and filled map – gauge – formatting – applications.	Uses of map visual – different functionalities of filled map, gauge and map visuals.	

WEEKLY REPORT

WEEK – 6 (From Dt 13-02-2023 to Dt 18-02-2023)

Scatter and bubble charts

A scatter chart shows the relationship between two numerical values. A bubble chart replaces data points with bubbles, with the bubble *size* representing a third data dimension.



Scatter charts are a great choice:

- To show relationships between two numerical values.
- To plot two groups of numbers as one series of x and y coordinates.
- To use instead of a line chart when you want to change the scale of the horizontal axis.
- To turn the horizontal axis into a logarithmic scale.
- To display worksheet data that includes pairs or grouped sets of values.

Tip

In a scatter chart, you can adjust the independent scales of the axes to reveal more information about the grouped values.

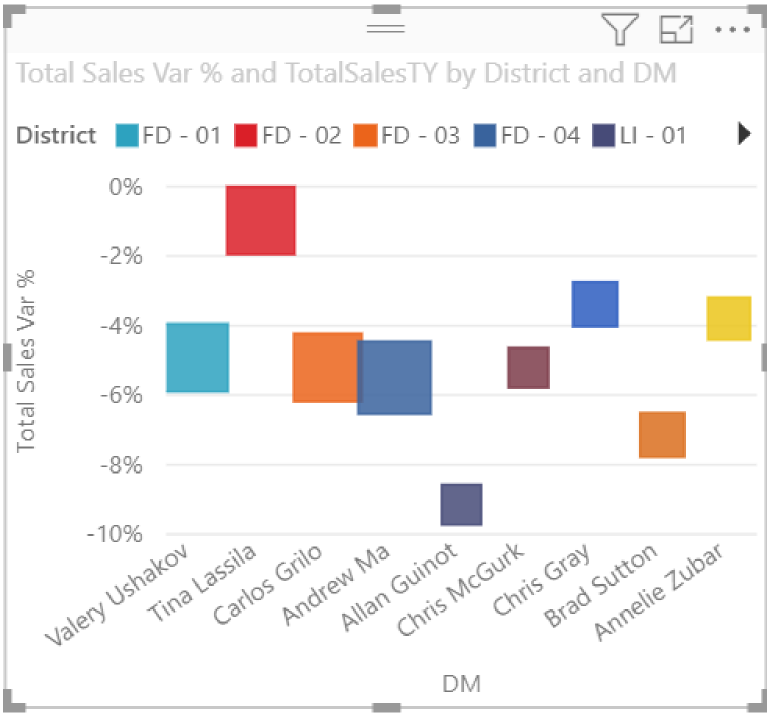
- To show patterns in large sets of data, for example by showing linear or non-linear trends, clusters, and outliers.
- To compare large numbers of data points without regard to time. The more data that you include in a scatter chart, the better the comparisons that you can make.

In addition to what scatter charts can do, bubble charts are a great choice:

- If your data has three data series that each contains a set of values.
- To present financial data. Different bubble sizes are useful to visually emphasize specific values.
- To use with quadrants.

Dot plot charts

A dot plot chart is similar to a bubble chart and scatter chart, but is instead used to plot categorical data along the horizontal axis.



They're a great choice if you want to include categorical data along the horizontal axis.

A filled, or *choropleth*, map uses shading or tinting or patterns to display how a value differs in proportion across a geography or region. You can quickly display relative differences with shading that ranges from light (less frequent/lower) to dark (more frequent/more).

Filled maps are a great choice:

- to display quantitative information on a map.
- to show spatial patterns and relationships.
- when your data is standardized.
- when working with socioeconomic data.
- when defined regions are important.
- to get an overview of the distribution across the geographic locations.

ACTIVITY LOG FOR THE SEVEN WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
20/02/2023 MONDAY	Seminar by students	Recap of what has been learnt so far	
21/02/2023 TUESDAY	Single card – multi card – their demonstration – formatting of cards	Several uses of single card and multi cards – applications – effectiveness of cards	
22/02/2023 WEDNESDAY	Key performance indicators – creation – setting a range for the KPI – indicating color formatting	Use of KPI visual – demonstration – report by students	
23/02/2023 THURSDAY	DAX – calculated columns vs calculated measures – variations – use of various DAX functions	Various functions of DAX – variations in calculated columns and calculated measures	
24/02/2023 FRIDAY	Row context – filter context – uses – differences – applications – writing queries	Using of row context and filter context to apply for data and create measures or columns	
25/02/2023 SATURDAY	Difference between SUM function and SUMX function – their uses	Use of aggregate functions – SUM and SUMX	

WEEKLY REPORT

WEEK – 7 (From Dt 20-02-2023 to Dt 25-03-2023)

Power BI Card visual

Sometimes we want to track our Power BI report or dashboard through a single number such as Total sales, total profits, total quantity, etc. For this, the **Power BI card visual** is the best choice. We can get this visual in Power BI, as it is available by default.

We can use the Power BI card visual for both Power BI Desktop and Power BI Service. It is useful to show a single numerical value or a metric value of a number. Also, we can use this to create a report editor or Q&A.

There are a few formatting options available in the Power BI Card, that we can see from the Format option.

It helps to make the card more attractive and eye-catching. We can change its text color, background color, data label, text size, design, and much more.

- **General:** Here we can set and format the X &Y position as well as height and width of the Power BI card visual.
- **Data label:** Here the data label is the numerical data which is visalize on the card chart. We can format this **data label color** (Black to Blue), **display units** (Auto to Thousand), **text size** (45 pt to 40 pt), **font family** (DIN to Comic sans MS), etc.

Benefits of a comprehensive KPI tool

KPI dashboards provide users with:

- A fast, easy solution to tracking KPIs and other business metrics.
- A unified view of data that improves visibility into company health.
- Customizable data visualization with performance and status indicators.

By building your KPI dashboard with the same tool or platform that you use to define your KPIs, you'll have everything you need in the same place. As you modify your KPI definitions, data sources, or targets, these updates will populate automatically into your dashboards—saving you valuable time and effort.

- **Background**: We can set a background color, choosing from the color chart to make our visual more attractive and set the transparency of the background color. Also, we can hide show or hide the background color by using the **toggling option** of it. For example, here we will set a background color and make it transparence.

By default, when we choose any numerical data field to create a card chart, it comes as SUM(total) of that data field. Now we will see how to customize this value or number field. For this, on the **formatting pane > Data label > Display unit.**

By building your KPI dashboard with the same tool or platform that you use to define your KPIs, you'll have everything you need in the same place. As you modify your KPI definitions, data sources, or targets, these updates will populate automatically into your dashboards—saving you valuable time and effort.

ACTIVITY LOG FOR THE EIGTH WEEK

Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
27/02/2023 MONDAY	Difference between CALENDAR and CALENDARAUTO function – their examples	Use of CALENDAR Function - various applications – creating a calendar	
28/02/2023 TUESDAY	Aggregate functions – DAX functions – its uses	Using various aggregate functions – calculated measures for various calculated values	
01/03/2023 WEDNESDAY	Applying CALENDAR Functions – use od DATE Functions	How to write DAX for CALENDAR Functions	
02/03/2023 THURSDAY	SUM – MIN – MAX – COUNT – AVERAGE – DISTINCTCOUNT - COUNTROWS	Using various DAX EXPRESSION FOR CALCULATED MEASURES	
03/03/2023 FRIDAY	TRIM – UPPER – LOWER – CONCATENATE – EXACT - & - text functions in DAX	Uses of Text Functions in power bi	
04/03/2023 SATURDAY	Seminar by students – documentation for report in Power BI	Recap of what has be learnt so far	

WEEKLY REPORT

WEEK – 8 (From Dt 27-02-2023 to Dt 04-03-2023)

CALENDAR FUNCTION Returns a table with one column of all dates between StartDate and EndDate.
Syntax
CALENDAR (<StartDate>, <EndDate>)

PARAMETER	ATTRIBUTES	DESCRIPTION
StartDate		The start date in datetime format.
EndDate		The end date in datetime format.

Return values
TABLE A table with a single column.

Returns a table with a single column named "Date" containing a contiguous set of dates. The range of dates is from the specified start date to the specified end date, inclusive of those two dates.

Remarks

The CALENDAR table is useful to create a Date table.
For compatibility with DAX time intelligence functions, it is a best practice to always include an entire year in a Date table.

» 2 related articles
» 1 related function

Examples

The following formula returns a table with dates between January 1st, 2005 and December 31st, 2015.

```
1  CALENDAR (
2      DATE ( 2005, 1, 1 ),
3      DATE ( 2015, 12, 31 ))
4  )
```

When we combine or summarize the numerical data, it is called "**Aggregation**." The output we get from this is called "**Aggregate**." So, the common aggregation functions are SUM, AVERAGE, MIN, MAX, COUNT, DISTINCTCOUNT, and so on. To use all these **aggregate functions**, we need some data that should be alpha-numerical. An aggregate function can be applied only for a numerical data set, but there is a situation where it also works for alphabetical data.

Function	**Description**
APPROXIMATEDISTINCTCOUNT	Returns an estimated count of unique values in a column.

Function	Description
AVERAGE	Returns the average (arithmetic mean) of all the numbers in a column.
AVERAGEA	Returns the average (arithmetic mean) of the values in a column.
AVERAGEX	Calculates the average (arithmetic mean) of a set of expressions evaluated over a table.
COUNT	Counts the number of rows in the specified column that contain non-blank values.
COUNTA	Counts the number of rows in the specified column that contain non-blank values.
COUNTAX	Counts non-blank results when evaluating the result of an expression over a table.
COUNTBLANK	Counts the number of blank cells in a column.
COUNTROWS	Counts the number of rows in the specified table, or in a table defined by an expression.
COUNTX	Counts the number of rows that contain a number or an expression that evaluates to a number, when evaluating an expression over a table.
DISTINCTCOUNT	Counts the number of distinct values in a column.
DISTINCTCOUNTNOBLANK	Counts the number of distinct values in a column.
MAX	Returns the largest numeric value in a column, or between two scalar expressions.
MAXA	Returns the largest value in a column.
MAXX	Evaluates an expression for each row of a table and returns the largest numeric value.
MIN	Returns the smallest numeric value in a column, or between two scalar expressions.
MINA	Returns the smallest value in a column, including any logical values and numbers represented as text.
MINX	Returns the smallest numeric value that results from evaluating an expression for each row of a table.
PRODUCT	Returns the product of the numbers in a column.
PRODUCTX	Returns the product of an expression evaluated for each row in a table.
SUM	Adds all the numbers in a column.
SUMX	Returns the sum of an expression evaluated for each row in a table.

ACTIVITY LOG FOR THE NINETH WEEK

Day & Date	Brief description of the dailyactivity	Learning Outcome	Person In-Charge Signature
06/03/2023 MONDAY	SWITCH – Uses – Applications – working – Query writing – IF – NESTED IF	Writing queries for logical values – difference between SWITCH & IF Functions	
07/03/2023 TUESDAY	PARENT – CHILD Functions - PATHLENGTH – PATHCONTAINS – PATHITEM – PATHITEMREVERSE – PATHLENGTH	Uses of PARENT CHILD Functions – Applications of those functions	
08/03/2023 WEDNESDAY	RELATED FUNCTION in DAX – CALCULATE FUNCTION – FILTER FUNCTION	Working of CALCULATE FUNCTION	
09/03/2023 THURSDAY	EXPLICIT, IMPLICIT FILTERS, KEEPFILTER FUNCTION IN DAX	Applying FILTERS in Query – Uses of KEEPFILTER Function	
10/03/2023 FRIDAY	SAMEPERIODLASTYEAR Function in DAX, DATEADD Function, USERELATIONSHIP Function	Uses od SAMEPERIODLASTYEAR Function – Uses of DATEADD Function	
11/03/2023 SATURDAY	ALL Function – use of ALL Function – ACTIVE connection – INACTIVE connection	Creating an ACTIVE connection – Uses of ALL Function	

WEEKLY REPORT

WEEK – 9 (From Dt 06-02-2023 to Dt 11-03-2023)

SWITCH

Evaluates an expression against a list of values and returns one of multiple possible result expressions.

DAXCopy

```
SWITCH(<expression>, <value>, <result>[, <value>, <result>]…[, <else>])
```

Term	Definition
expression	Any DAX expression that returns a single scalar value, where the expression is to be evaluated multiple times (for each row/context).
value	A constant value to be matched with the results of expression.
result	Any scalar expression to be evaluated if the results of expression match the corresponding value.
else	Any scalar expression to be evaluated if the result of expression doesn't match any of the value arguments.

A scalar value coming from one of the result expressions, if there was a match with value, or from the else expression, if there was no match with any value.All result expressions and the else expression must be of the same data type.The following example creates a calculated column of month names.

DAXCopy

```
= SWITCH([Month], 1, "January", 2, "February", 3, "March", 4, "April"
               , 5, "May", 6, "June", 7, "July", 8, "August"
               , 9, "September", 10, "October", 11, "November", 12, "December"
               , "Unknown month number" )
```

Parent and Child functions

These functions manage data that is presented as parent/child hierarchies. To learn more, see Understanding functions for Parent-Child Hierarchies in DAX.

Function	Description
PATH	Returns a delimited text string with the identifiers of all the parents of the current identifier.
PATHCONTAINS	Returns TRUE if the specified item exists within the specified path.
PATHITEM	Returns the item at the specified position from a string resulting from evaluation of a PATH function.
PATHITEMREVERSE	Returns the item at the specified position from a string resulting from evaluation of a PATH function.
PATHLENGTH	Returns the number of parents to the specified item in a given PATH result, including s

RELATED - Returns a related value from another table.

DAXCopy

```
RELATED(<column>)
```

A single value that is related to the current row.

Remarks

- The RELATED function requires that a relationship exists between the current table and the table with related information. You specify the column that contains the data that you want, and the function follows an existing many-to-one relationship to fetch the value from the specified column in the related table. If a relationship does not exist, you must create a relationship.
- When the RELATED function performs a lookup, it examines all values in the specified table regardless of any filters that may have been applied.
- The RELATED function needs a row context; therefore, it can only be used in calculated column expression, where the current row context is unambiguous, or as a nested function in an expression that uses a table scanning function. A table scanning function, such as SUMX, gets the value of the current row value and then scans another table for instances of that value.
- The RELATED function cannot be used to fetch a column across a limited relationship.

Explicit measures = more flexibility!

Writing measures in an explicit way, using DAX language, requires more time and effort in the beginning since you need to do some manual work. But, you will bear the fruits later, believe me.

ACTIVITY LOG FOR THE TENTH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
13/03/2023 MONDAY	Recalling USERALATIONSHIP Function	Use of USERELATIONSHIP Function in DAX – Applications.	
14/03/2023 TUESDAY	ALLSELECTED – ALLEXCEPT Function in DAX – LOOKUP Value – RANKX Function in POWER BI	Uses of Various Functions in DAX – applications – Writing a query – Dos and don' t s	
15/03/2023 WEDNESDAY	YTD – MTD – QTD – SUMMARIZE – SUMMARIZED COLUMN	How to use SUMMARIZE Function – Difference between SUMMARIZE & SUMMARIZED COLUMN	
16/03/2023 THURSDAY	CALCULATE VS CALCULATED TABLE – MAX – MAXX – MAXA – IFERROR	DIFFERENCE IN MAX – MAXA – MAXX FUNCTIONS – Use of IFERROR Function	
17/03/2023 FRIDAY	Variables in power BI – DATEIFF – ISBLANK Function in DAX	Applications od DATEIFF Function - Use of ISBLANK Function	
18/03/2023 SATURDAY	USERNAME() & USERPRINCIPALNAME() – COUNT() – COUNTA() – COUNTX	Getting to know about POWER SERVICES – use of USERNAME() Function	

WEEKLY REPORT

WEEK – 10 (From Dt 13-03-2023 to Dt 18-03-2023)

USERELATIONSHIP

Specifies the relationship to be used in a specific calculation as the one that exists between columnName1 and columnName2.

USERELATIONSHIP(<columnName1>,<columnName2>)

Term	**Definition**
columnName1	The name of an existing column, using standard DAX syntax and fully qualified, that usually represents the many side of the relationship to be used; if the arguments are given in reverse order the function will swap them before using them. This argument cannot be an expression.
columnName2	The name of an existing column, using standard DAX syntax and fully qualified, that usually represents the one side or lookup side of the relationship to be used; if the arguments are given in reverse order the function will swap them before using them. This argument cannot be an expression.

The function returns no value; the function only enables the indicated relationship for the duration of the calculation.

Remarks

- USERELATIONSHIP can only be used in functions that take a filter as an argument, for example: CALCULATE, CALCULATETABLE, CLOSINGBALANCEMONTH, CLOSINGBALANCEQUARTER, CLOSINGBALANCEYEAR, OPENINGBALANCEMONTH, OPENINGBALANCEQUARTER, OPENINGBALANCEYEAR, TOTALMTD, TOTALQTD and TOTALYTD functions.
- USERELATIONSHIP cannot be used when row level security is defined for the table in which the measure is included. For example, CALCULATE(SUM([SalesAmount]), USERELATIONSHIP(FactInternetSales[CustomerKey], DimCustomer[CustomerKey])) will return an error if row level security is defined for DimCustomer.
- USERELATIONSHIP uses existing relationships in the model, identifying relationships by their ending point columns.
- In USERELATIONSHIP, the status of a relationship is not important; that is, whether the relationship is active or not does not affect the usage of the function. Even if the relationship is inactive, it will be used and overrides any other active relationships that might be present in the model but not mentioned in the function arguments.
- An error is returned if any of the columns named as an argument is not part of a relationship or the arguments belong to different relationships.
- If multiple relationships are needed to join table A to table B in a calculation, each relationship must be indicated in a different USERELATIONSHIP function.
- If CALCULATE expressions are nested, and more than one CALCULATE expression contains a USERELATIONSHIP function, then the innermost USERELATIONSHIP is the one that prevails in case of a conflict or ambiguity.

- Up to 10 USERELATIONSHIP functions can be nested; however, your expression might have a deeper level of nesting, ie. the following sample expression is nested 3 levels deep but only 2 for USERELATIONSHIP: =CALCULATE(CALCULATE(CALCULATE(<anyExpression>, USERELATIONSHIP(t1[colA], t2[colB])), t99[colZ]=999), USERELATIONSHIP(t1[colA], t2[colA])).
- For 1-to-1 relationships, USERELATIONSHIP will only activate the relationship in one direction. In particular, filters will only be able to flow from *columnName2*'s table to *columnName1*'s table. If bi-directional cross-filtering is desired, two USERELATIONSHIPs with opposite directionality can be used in the same calculation. For example, CALCULATE(..., USERELATIONSHIP(T1[K], T2[K]), USERELATIONSHIP(T2[K], T1[K])).

YTD will display all fields from January to selected date, QTD should display only those fields which are applicable for that quarter and MTD should display only those fields which are applicable for that month.Sales should come as blank if any field sales is not present in QTD and MTD Sales.

ALLSELECTED

Removes context filters from columns and rows in the current query, while retaining all other context filters or explicit filters.

The ALLSELECTED function gets the context that represents all rows and columns in the query, while keeping explicit filters and contexts other than row and column filters. This function can be used to obtain visual totals in queries.

Syntax

DAXCopy

ALLSELECTED([<tableName> | <columnName>[, <columnName>[, <columnName>[,…]]]])

Term	Definition
tableName	The name of an existing table, using standard DAX syntax. This parameter cannot be an expression. This parameter is optional.
columnName	The name of an existing column using standard DAX syntax, usually fully qualified. It cannot be an expression. This parameter is optional.

Return value

The context of the query without any column and row filters.

Remarks

- If there is one argument, the argument is either *tableName* or *columnName*. If there is more than one argument, they must be columns from the same table.
- This function is different from ALL() because it retains all filters explicitly set within the query, and it retains all context filters other than row and column filters.
- This function is not supported for use in DirectQuery mode when used in calculated columns or row-level security (RLS) rules.

ACTIVITY LOG FOR THE ELEVENTH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
20/03/2023 MONDAY	Seminar by students	Recap of what has been learnt so far	
21/03/2023 TUESDAY	PARALLEL PERIOD – as a FILTER in CALCULATED Function – DATEVALUE Function in DAX	Understanding PARALLELPERIOD – DATEVALUE Functions in power bi	
22/03/2023 WEDNESDAY	UGADI HOLIDAY	------	
23/03/2023 THURSDAY	Communications and soft skills - How to keep the conversation flowing – knowing how to answer Strengths and weakness – portraying skills through communicating	Things to keep in mind while attending an interview – maintaining a proper etiquette	
24/03/2023 FRIDAY	Communications and soft skills class – JAM Sessions	Improving communication skills – being able to talk about a topic in a correct manner – facing audience	
25/03/2023 SATURDAY	Seminar by students	Recap of what has been learnt so far	

WEEKLY REPORT

WEEK – 11 (From Dt 20-03-2023 to Dt 25-04-2023)

PARALLELPERIOD

Returns a table that contains a column of dates that represents a period parallel to the dates in the specified **dates** column, in the current context, with the dates shifted a number of intervals either forward in time or back in time.

DAXCopy
PARALLELPERIOD(<dates>,<number_of_intervals>,<interval>)

Term	Definition
dates	A column that contains dates.
number_of_intervals	An integer that specifies the number of intervals to add to or subtract from the dates.
interval	The interval by which to shift the dates. The value for interval can be one of the following: year, quarter, month.

A table containing a single column of date values.

Remarks

- This function takes the current set of dates in the column specified by **dates**, shifts the first date and the last date the specified number of intervals, and then returns all contiguous dates between the two shifted dates. If the interval is a partial range of month, quarter, or year then any partial months in the result are also filled out to complete the entire interval.
- The **dates** argument can be any of the following:
- A reference to a date/time column,
- A table expression that returns a single column of date/time values,
- A Boolean expression that defines a single-column table of date/time values.
- Constraints on Boolean expressions are described in the topic, CALCULATE function.
- If the number specified for **number_of_intervals** is positive, the dates in **dates** are moved forward in time; if the number is negative, the dates in **dates** are shifted back in time.
- The **interval** parameter is an enumeration, not a set of strings; therefore values should not be enclosed in quotation marks. Also, the values: year, quarter, month should be spelled in full when using them.
- The result table includes only dates that appear in the values of the underlying table column.
- The PARALLELPERIOD function is similar to the DATEADD function except that PARALLELPERIOD always returns full periods at the given granularity level instead of the partial periods that DATEADD returns. For example, if you have a selection of dates that starts at June 10 and finishes at June 21 of the same year, and you want to shift

that selection forward by one month then the PARALLELPERIOD function will return all dates from the next month (July 1 to July 31); however, if DATEADD is used instead, then the result will include only dates from July 10 to July 21.

- This function is not supported for use in DirectQuery mode when used in calculated columns or row-level security (RLS) rules.

Communication skills may take a lifetime to master—if indeed anyone can ever claim to have mastered them. There are, however, many things that you can do fairly easily to improve your communication skills and ensure that you are able to transmit and receive information effectively.

The Importance of Good Communication Skills

Developing your communication skills can help all aspects of your life, from your professional life to social gatherings and everything in between.

The ability to communicate information accurately, clearly and as intended, is a vital life skill and something that should not be overlooked. It's never too late to work on your communication skills and by doing so, you may well find that you improve your quality of life.

Communication skills are needed in almost all aspects of life:

- **Professionally, if you are applying for jobs or looking for a promotion with your current employer, you will almost certainly need to demonstrate good communication skills.**

  Communication skills are needed to speak appropriately with a wide variety of people whilst maintaining good eye contact, demonstrate a varied vocabulary and tailor your language to your audience, listen effectively, present your ideas appropriately, write clearly and concisely, and work well in a group. Many of these are essential skills that most employers seek.

ACTIVITY LOG FOR THE TWELVETH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
27/03/2023 MONDAY	Seminar by students	Exchanging ideas for better understanding of a topic	
28/03/2023 TUESDAY	Percentage chart - 100% stacked bar chart – 100% stacked column chart	Uses of percentage column and bar charts – its formatting	
29/03/2023 WEDNESDAY	Seminar by students – clarification of doubts imposed by students – understanding the problem	Interacting with one another and team playing - brainstorming	
30/03/2023 THURSDAY	SRI RAMA NAVAMI HOLIDAY	------	
31/03/2023 FRIDAY	AVERAGE, AVERAGEA, AVERAGEX, MIN, MINX, MINA Functions in DAX.	Understanding AVERAGE &AVERAGEX Functions in DAX	
01/04/2023 SATURDAY	Communications and Soft Skills class – common interview questions – demo answers – etiquettes to be followed	How to answer a question – dos and don 't s while attending an interview	

WEEKLY REPORT

WEEK – 12 (From Dt 27-03-2023 to Dt 01-04-2023)

DAX (Data Analysis Expression):

Data Analysis Expressions (DAX) is a programming language that is used throughout Microsoft Power BI for creating calculated columns, measures, and custom tables. It is a collection of functions, operators, and constants that can be used in a formula, or expression, to calculate and return one or more values.

DAX is a formula language. You can use DAX to define custom calculations for Calculated Columns and for Measures (also known as calculated fields). DAX includes some of the functions used in Excel formulas, and additional functions designed to work with relational data and perform dynamic aggregation.

DAX COLUMNS:

Calculated columns use Data Analysis Expressions (DAX) formulas to define a column's values. This tool is useful for anything from putting together text values from a couple of different columns to calculating a numeric value from other values.

DAX MEASURES:

The way of defining calculations in a DAX model, which helps us to calculate values based on each row. But rather, it gives us aggregate values from multiple rows from a table. Measures and calculated columns both use DAX expressions. The difference is the context of evaluation. A measure is evaluated in the context of the cell evaluated in a report or in a DAX query, whereas a calculated column is computed at the row level within the table it belongs to.

FILTER CONTEXT:

The filter context is the set of filters applied to the data model before the evaluation of a DAX expression starts. When you use a measure in a pivot table, for example, it produces different results for each cell because the same expression is evaluated over a different subset of the data.

ROW CONTEXT:

Row context can be thought of as "the current row.” If you have created a calculated column, the row context consists of the values in each individual row and values in columns that are related to the current row.

SUM & SUMX:

SUMX is an iterator function and takes a different approach. Unlike SUM, SUMX is capable of performing row-by-row calculations and iterates through every row of a specified table to complete the calculation. SUMX then adds all the row-wise results of the iterations of the given expression. Sum and SumX *both are functions calculating aggregation*.

ACTIVITY LOG FOR THE THIRTEENTH WEEK

Day & Date	**Brief description of the daily activity**	**Learning Outcome**	**Person In-Charge Signature**
03/04/2023 MONDAY	Related functions in DAX – uses of various RELATED FUNCTIONS	Use of various RELATED Functions in DAX	
04/04/2023 TUESDAY	MAHAVEER JAYANTHI HOLIDAY	------	
05/04/2023 WEDNESDAY	Parent child functions in DAX – conditions applied for parent child functions	Use of various parent child functions	
06/04/2023 THURSDAY	Aggregate functions in DAX – importance of aggregate function.	Use of aggregate functions – how to apply DAX to columns or measures	
07/04/2023 FRIDAY	GOOD FRIDAY HOLIDAY	------	
08/04/2023 SATURDAY	DAX calculation types – calculated columns and calculated measures	Calculated columns vs calculated measures – advantages - disadvantages	

WEEKLY REPORT

WEEK – 13 (From Dt 03-04-2023 to Dt 08-04-2023)

SAME PERIOD LAST YEAR FUCNTION AND DATE ADD FUNCTION: -
returns a table that contains a column of dates shifted one year back in time from the dates in the specified **dates** column, in the current context.

SAMEPERIODLASTYEAR(<dates>)

- Dates – A column that contains dates.

SAMEPERIODLASTYEAR returns a single column table of dates values. SAMEPERIODLASTYEAR is specific to a year interval where a syntax for the DATEADD and PARALLELPERIOD include other intervals such as quarter, month day.

DATEADD returns a table that contains a column of dates, shifted either forward or backwards in time by the specified number of intervals from the dates in the current context.

DATEADD(<dates>,<number_of_intervals>,<interval>)

- Dates – A column that contains dates.
- Number_of_intervals – an integer that specifies the number of intervals to add to or subtract from the dates.
- Interval – the interval by which to shift the dates. The value for interval can be one of the following, year, quarter, month or day.

We can see that DATEADD look at a set of dates have a number of intervals that it looks backwards or forward and goes backwards or forward based on a set interval of a year, quarter or month.

All Functions in DAX:-

Returns all the rows in a table, or all the values in a column, ignoring any filters that might have been applied. This function is useful for clearing filters and creating calculations on all the rows in a table

Use Relationship in DAX:-
USERELATIONSHIP is a CALCULATE modifier, which instructs CALCULATE to temporarily activate the relationship. When CALCULATE has computed its result, the default relationship becomes active again.

All except and lookup value functions:-
Returns all the rows in a table except for those rows that are affected by the specified column filters.

A value that LOOKUP searches for in an array. The lookup value argument can be a number, text, a logical value, or a name or reference that refers to a value.

RANK X FUNCTION:-
RANKX is a type of function in Power BI. It is a built-in function termed a sorting function, which is used extensively in sorting the data in various conditions. The syntax for this function is as follows: RANKX(<table>, <expression>, <value>, <order>, <ties>).

ACTIVITY LOG FOR THE FOUTEENTH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
10/04/2023 MONDAY	Presentations of sample datasets by students	Being able to identify the data – understanding the report	
11/04/2023 TUESDAY	Working on sample dataset – creating visuals – writing DAX queries	Understanding how to implement various DAX queries to obtain the desired outcome	
12/04/2023 WEDNESDAY	Seminar by students – clarification of doubts imposed by students – understanding the problem	Interacting with one another and team playing - brainstorming	
13/04/2023 THURSDAY	Presentations of sample datasets by students	Understanding various types of datasets and observing the visuals used – observation – questioning - understanding	
14/04/2023 FRIDAY	Presentations of sample datasets by students	Being able to identify the data – understanding the report	
15/04/2023 SATURDAY	Seminar by students	Attempting to find alternative solutions - brainstorming	

WEEKLY REPORT

WEEK – 14 (From Dt 10-04-2023 to Dt 15-04-2023)

WEEK – 14 (From Dt 10-04-2023 to Dt 15-04-2023)

Total Year to Date: Evaluates the year-to-date value of the expression in the current context.

Total Month to Date: Evaluates the month to date value of the expression in the current context.

Total Quarter to Date: Evaluates the quarter-to-date value of the expression in the current context.

Syntax:(Sum (Expression, Dates))

Summarize: It is used to extract the data and creates the table. We should pass the table name in summarize column. By default, it does not have any filter function.

Summarize Column: It is used to extract the data and creates the table. DAX engine automatically finds the table name so we do not need to pass the table name explicitly.

Calculate Column: It returns a scaler value. Evaluates an expression in context modify by filter.

Calculate table: It returns a table. Evaluates a table expression in a context modify by filter.

MAX: Returns the numeric largest value or largest string value in a column.

Syntax: Max (Column Name)

MAXX: Returns the numeric largest value or largest string value in a column that result from evaluating and expression from each row of a table.

Syntax: MAXX(Table, Expression)

MAXA: Returns the largest numerical value in a column does not ignores the logical value.

Syntax: MAXA(Column Name)

Variable in DAX: Variables are declared by using “Var” Keyword and it should end with return keyword to break the variable flow end. Variables can be used in measures, Columns and table.

Date Iff: return the no of intervals boundaries between the two dates.

Syntax: Dateiff(Date1, Date2, Interval)

Date Value: Converts a date into text format to dates in dates format.

Syntax: Date value(Date Column)

Parallel Period: Returns the parallel period of the dates by the given set of dates and a specified internal.

Syntax: Parallel Period(Date, no of Intervals, Interval)

ACTIVITY LOG FOR THE FIFTEENTH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
17/04/2023 MONDAY	Working on the project – searching datasets – understanding data	Recalling various concepts that have been taught	
18/04/2023 TUESDAY	Searching through various platforms – transforming data – using various visuals	Learning to understand the data and applying suitable visual to represent the outcome	
19/04/2023 WEDNESDAY	Working on the project – understanding various visuals for presenting data	Using various alternatives – dos and donts for applying a visual to project the outcome	
20/04/2023 THURSDAY	ANNUAL DAY CELEBRATIONS	------	
21/04/2023 FRIDAY	ANNUAL DAY CELEBRATIONS	------	
22/04/2023 SATURDAY	RAMZAN HOLIDAY	------	

WEEKLY REPORT

WEEK – 15 (From Dt 17-04-2023 to Dt 22-04-2023)

CALCULATE

Evaluates an expression in a modified filter context.There's also the **CALCULATETABLE** function. It performs exactly the same functionality, except it modifies the **filter context** applied to an expression that returns a table object.

DAXCopy
CALCULATE(<expression>[, <filter1> [, <filter2> [, …]]])

The expression used as the first parameter is essentially the same as a measure.

When there are multiple filters, they can be evaluated by using the AND (&&) logical operator, meaning all conditions must be TRUE, or by the OR (||) logical operator, meaning either condition can be true.

Boolean filter expressions

A Boolean expression filter is an expression that evaluates to TRUE or FALSE. There are several rules that they must abide by:

- They can reference columns from a single table.
- They cannot reference measures.
- They cannot use a nested CALCULATE function.

Beginning with the September 2021 release of Power BI Desktop, the following also apply:

- They cannot use functions that scan or return a table unless they are passed as arguments to aggregation functions.
- They can contain an aggregation function that returns a scalar value. For example,
  DAXCopy
  ```
  Total sales on the last selected date =
  CALCULATE (
      SUM ( Sales[Sales Amount] ),
      'Sales'[OrderDateKey] = MAX ( 'Sales'[OrderDateKey] )
  )
  ```

Table filter expression

A table expression filter applies a table object as a filter. It could be a reference to a model table, but more likely it's a function that returns a table object. You can use the FILTER function to apply complex filter conditions, including those that cannot be defined by a Boolean filter expression.

Filter modifier functions

Filter modifier functions allow you to do more than simply add filters. They provide you with additional control when modifying filter context.

Function	Purpose
REMOVEFILTERS	Remove all filters, or filters from one or more columns of a table, or from all columns of a single table.
ALL [1], ALLEXCEPT, ALLNOBLANKROW	Remove filters from one or more columns, or from all columns of a single table.
KEEPFILTERS	Add filter without removing existing filters on the same columns.
USERELATIONSHIP	Engage an inactive relationship between related columns, in which case the active relationship will automatically become inactive.
CROSSFILTER	Modify filter direction (from both to single, or from single to both) or disable a relationship.

[1] The ALL function and its variants behave as both filter modifiers and as functions that return table objects. If the REMOVEFILTERS function is supported by your tool, it's better to use it to remove filters.

Return value

The value that is the result of the expression.

Remarks

- When filter expressions are provided, the CALCULATE function modifies the filter context to evaluate the expression. For each filter expression, there are two possible standard outcomes when the filter expression is not wrapped in the KEEPFILTERS function:
- If the columns (or tables) aren't in the filter context, then new filters will be added to the filter context to evaluate the expression.
- If the columns (or tables) are already in the filter context, the existing filters will be overwritten by the new filters to evaluate the CALCULATE expression.
- The CALCULATE function used without filters achieves a specific requirement. It transitions row context to filter context. It's required when an expression (not a model measure) that summarizes model data needs to be evaluated in row context. This scenario can happen in a calculated column formula or when an expression in an iterator function is evaluated. Note that when a model measure is used in row context, context transition is automatic.
- This function is not supported for use in DirectQuery mode when used in calculated columns or row-level security (RLS) rules.

ACTIVITY LOG FOR THE SIXTEEENTH WEEK

Day & Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
24/04/2023 MONDAY	Presentations of sample datasets by students	Being able to identify the data – understanding the report	
25/04/2023 TUESDAY	Working on sample dataset – creating visuals – writing DAX queries	Understanding how to implement various DAX queries to obtain the desired outcome	
26/04/2023 WEDNESDAY	Seminar by students – clarification of doubts imposed by students – understanding the problem	Interacting with one another and team playing - brainstorming	
27/04/2023 THURSDAY	Presentations of sample datasets by students	Understanding various types of datasets and observing the visuals used – observation – questioning - understanding	
28/04/2023 FRIDAY	Presentations of sample datasets by students	Being able to identify the data – understanding the report	
29/04/2023 SATURDAY	Seminar by students	Attempting to find alternative solutions - brainstorming	

WEEKLY REPORT

WEEK – 16 (From Dt 24-04-2023 to Dt 29-04-2023)

Visualize in Power BI

- Power BI datasets represent a source of data that's ready for reporting and visualization. You can create Power BI datasets in the following ways:
- Connect to an existing data model that isn't hosted in Power BI.
- Upload a Power BI Desktop file that contains a model.
- Upload an Excel workbook that contains one or more Excel tables
- a workbook data model, or upload a comma-separated values (CSV) file.
- Use the Power BI service to create a dataset.
- Use the Power BI service to create a streaming or hybrid streaming of data set.
- A Power BI visualization is constructed using data from your datasets. If you're interested in seeing behind-the-scenes, Power BI lets you display the data that is being used to create the visual.
- When you select Show as a table, Power BI displays the data as a table below (or next to) the visualization.
- You can also export the data that is being used to create the visualization as an .xlsx or.csv file and view it in Excel.

- A Power BI report is the "Power BI way" of showing findings and insights from a dataset.
- This is one of the things that differentiate Power BI from Excel where data is presented in rows and columns right off the bat.
- Simply, Power BI reports are visuals These reports came from a dataset. The visuals could then be pinned on dashboards.
- Before you can create any reports in Power BI, you need a dataset and a blank report canvas. There are three primary sections you have to know:
- Canvas, Fields pane, Visualizations Pane.
- Obviously, the canvas is the empty white space at the center where the visuals will be made.
- The fields pane contains the different fields of your dataset (think of it as the columns of
- your dataset).
- The visualizations pane is where you can edit and modify your visualizations like the type of visualization, the format, and specific options for the values of your visual.
- If you have tried creating charts and graphs for your data in Excel, then creating
- reports in Power BI is even easier.
- Select the fields first then visualizations after.
- Or if you have the specific visualization in mind, select the type of visual first then the fields after.

ACTIVITY LOG FOR THE SEVENTEENTH WEEK

Day & Date	**Brief description of the daily activity**	**Learning Outcome**	**Person In-Charge Signature**
01/05/2023 MONDAY	Working on the project – searching datasets – understanding data	Recalling various concepts that have been taught	
02/05/2023 TUESDAY	Searching through various platforms – transforming data – using various visuals	Learning to understand the data and applying suitable visual to represent the outcome	
03/05/2023 WEDNESDAY	Working on the project – understanding various visuals for presenting data	Using various alternatives – dos and don’ts for applying a visual to project the outcome	
04/05/2023 THURSDAY	Searching through various platforms – transforming data – using various visuals	Learning to understand the data and applying suitable visual to represent the outcome	
05/05/2023 FRIDAY	Working on the project – understanding various visuals for presenting data	Using various alternatives – dos and don’ts for applying a visual to project the outcome	
06/05/2023 TO 10/05/2023	Preparing the documentation for the project that has been completed	------	

WEEKLY REPORT

WEEK – 17 (From Dt 01-05-2023 to Dt 10-05-2023)

Worked on various datasets. Searched for online sources to obtain datasets tried out different online sources to obtain data sets. Imported the data into power bi and tried various visuals for the given data. Discussing how to apply various visuals to obtain certain results for a visual. Created meaningful dashboard that can be used for drawing various conclusion and analyzing the pattern of the numbers in the data set. Prepared documentation for the project and proceeded to enter the log for the required period of time. Cross checked each and every topic mention in the document regarding the project. Included the roles and responsibilities that we have followed during the internship period. Mentioned various managerial, communication, technical and team management skills that we have acquired during the period.

Overview Of the Internship Program

Work environment I have experienced

A unique teaching style to help aspiring learners to master the art of managing data and creating powerful dashboards to make intelligent business decisions. Working together to turn your unrelated sources of data into clear, visually immersive, and interactive insights.

Individually getting data from a wide range of systems in the cloud and create dashboards that will track the metrics they care the most. With this technology, enterprises can see their business performance more closely and get immediate results with real-time dashboards available for every device.

All the topics and units are broken down in an easy to learn manner, making the course extremely enjoyable and our friendly instructors delight all our learners to use Power BI successfully. The trainers will help you understand how to use Power BI to analyze data, identify trends and make better decisions. They helped learn how to create reports, dashboards, and interactive visuals with Power BI. Helped us get the most out of Power BI and make it work for your business.

Interactive and highly engaging BI training course helps candidates across departments communicate performance-related data in a visually understandable manner.

Exploring the different types of testing, preparing, and presenting data quickly and easily. Learn how to turn data into powerful insights that direct and drive informed business decisions - giving your business the edge over competitors.

Learnt about DAX data computation, creating tables, binned tables, data drill down and drill up, Power Q&A, and many more advanced topics.

learn how to create and customize reports, work with advanced analytics, and more.

Real time technical skills I have acquired

The Power BI Developer will have an advantage if they are well-versed in the Microsoft Business Intelligence stack, which includes Power Pivot, SSRS, SSIS, and SSAS. Since most companies use Microsoft services and goods, a Power BI Developer with a good scope of knowledge is an asset to the company.

Additionally, we can connect to cloud databases like Amazon Redshift, cloud applications like Salesforce, and warehouses. Power BI can connect to any type of data we can imagine. In addition, Power BI has a Web Data Connector that can connect any desired data source by pulling API directly from the web.

Business users with access to Power BI can quickly and easily find valuable data in their sizable Hadoop data sets. With this software, users no longer need to be familiar with query languages, making it easier for stakeholders to interact with big data.

Power BI is a business intelligence tool with excellent prospects for the foreseeable future. Its benefits and uses are significantly higher than any doubt that it might go off the market.

Data analysis involves the process of transforming raw data into useful information. It is a process applied to various industries. Power BI can connect to such sources as Google Analytics.

It is easier to understand all the different kinds of reports available in the product as we understand the architecture and components of Power BI.
We can create our data models and business rules.

- Proficient with using various visualization tools for this.
- Interactive visualizations of data from various sources. You'll gain
- Familiarity with the various Power BI visualizations available as a result
- Find the KPIs with the appropriate objectives
- Use current and historical data for better decision making
- Technical publications can be derived from business requirements
- Data models that are multi-dimensional
- Create data documentation about parameters, algorithms and models.
- Perform detailed analysis on deployed and tested PowerBI scripts.
- Showcase our advanced user skills in a professional portfolio.

Managerial skills I have acquired

Planning is a vital aspect within an organization. Process of formulating a set of actions or one or more strategies to pursue and achieve certain goals or objectives with the available resources. Process includes identifying and setting achievable goals, developing necessary strategies, and outlining the tasks and schedules on how to achieve the set goals. Information shared throughout a team, ensuring that the group acts as a unified workforce. Flow of information within the organization, whether formal or informal, verbal or written, vertical or horizontal, and it facilitates the smooth functioning of the organization. Making proper and right decisions results in the success of the organization, while poor or bad decisions may lead to failure or poor performance.

negotiate the terms of a contract or something discussed

Avoiding wastage of time, optimizes productivity, and ensures responsibility and accountability on the part. Ability to tackle and solve the frequent problems that can arise in a typical workday. Involves identifying a certain problem or situation and then finding the best way to handle the problem and get the best solution.

creating a safe and positive environment

Exhibit flexibility in their approaches to fast-paced work environments where change is inevitable and welcome. This includes being open to updating processes and applications to meet current industry trends and taking on additional tasks or responsibilities in order to accommodate schedule changes, short- or long-term work overloads, or other needs.

Demonstrating time management skills because it shows that even when handling multiple projects or responsibilities, work gets done, deadlines are met, and everyone's ducks are in neat little rows. Creating a schedule, stick to it, and maintain a perfect work-life balance, they're looking at a prime candidate.

listening supersedes speaking. We usually hear of "good" listening being a key takeaway, but proper listening should be more than good. It should be *active*. Active listening requires the "listener" to concentrate on what's being said rather than listening for the sake of responding. glazing over larger concerns embedded into an team's complaint or inquiry.

Motivation helps bring forth a desired behavior or response from certain stakeholders depending on characteristics such as company and team culture, team personalities, and more. A collection of abilities that include things such as business planning, decision-making, problem-solving, communication, delegation, and time management.

How I could improve my communication skills

Being able to convey complex, technical concepts to others – not just through their Power BI dashboards and reports, but via face-to-face communication, too. Keeping the communication concise without compromising on the importance of it. This applies to both written and verbal communication. For written communication, proofread, and for verbal communication, practice saying only what is important to the conversation.

Gauge what type of communication they are going to understand. For example, if you are communicating with a colleague or a senior, obviously informal language should not be used. Also, if you use acronyms, you cannot assume that the other person will immediately understand. So, know your listener.

The language you use in your communication should be assertive and active. This form of language instantly grabs the attention of the listener or reader. They will latch on to your every word and the right message will be passed on.

Keeping a positive body language like an open stance and eye contact. This is subconsciously read by the other person, and their body language also becomes positive.

Proofread what you have written once or twice before sending. One tip is that do not proofread immediately after writing. It's harder to spot errors. Take a small break, give rest to your eyes, and then proofread.

Making sure that you are in the right frame of mind. Tiredness, frustration, sadness, and anger, among other range of emotions, can hamper what we want to communicate. Just making sure we are positive or at least neutral.

Directly communicate with the person we mean to. In many organizations, communication channels are created with many needless people passing on the messages.

Communication is something that has a substantial impact on our personal and professional life. It has to be taken seriously. And always remembering some of the most successful and happy people in life are great communicators.

Describe how could you could enhance your abilities in group discussions, participationin teams, contribution as a team member, leading a team/a

Problem-solving skills: Candidates should be able to evaluate complex situations, analyze data, and reach intelligent conclusions in order to deliver effective solutions.

Good attention to detail: As they say, the devil is in the detail. Candidates will be expected to work with large volumes of data, and they must have a keen eye for smaller details, errors, and flaws in the data to make sure your business is working with only the cleanest, most accurate data.

"No man is an island". So, the saying goes. Increasingly in the workplace, we all have to work with others in order to complete a project. Be, it working in a team, or dealing with clients or suppliers, interpersonal abilities is a definite advantage and something employers always look for. The ability to build relationships with those around you under any circumstances, and the ability to inspire them to do what needs to be done is essential.

Theoretically, when someone is offered a job, there is a job description included in the contract. In reality however, employees are not expected to stick to only what is under their job description. On, the contrary, they are expected to get involved in other areas of the business, understand all the different steps, and offer help where necessary. At the end of the day, employers look for someone willing to try out different things, and wear multiple hats at the same time, deal with different projects and individuals, and provide more than one sole contribution at a time to the company.

Being self-confident exudes an aura that can convince those you work for (or with) that you know what you are doing. If you do not believe in yourself, your skills and abilities, then you cannot expect anyone else to believe in you. You need to be confident with yourself and ensure everyone sees you as someone that has the ability to pull through whatever situation comes your way.

Public speaking is a very crucial skill to have, which requires a lot of self-confidence, practice, and analyzing of your audience. Even though it comes naturally to some people, it is definitely a skill that can be acquired, and it is a skill sought after by employers.

No matter how much you believe you are right about something, or that it may be more useful for your colleague to know exactly what you think, realize that not everyone reacts the same way to different styles of confrontation. It is important for individuals to know how and when to deal with various issues that may crop up in the working environment, whether they are dealing with clients, colleagues, or supervisors.

Integrity and well-founded moral values should be highly-respected in the work place. Even though many scandals appear with black sheep here and there, it is essential for employees to maintain their values and integrity at all costs. Honesty and sticking to your values will definitely repay in the long run. An untainted reputation after all is what will help you move up the career ladder. Employers, always look for employees that are passionate about what they do and are very committed to their assignments. They need to be assured that their employees will keep at a problem until it is solved, and they will do what is necessary to complete all tasks.

Arriving at work on time and willingness to work and take responsibility are basic indicators of an employee's commitment. These factors can show whether an employee is cut out for a specific role.

If a person demonstrates an attitude that is appreciative of feedback, it can be deduced that he/she is willing to learn. Irrespective of age and experience, everyone is constantly learning at the workplace, and one should always remain open to new information that can enhance their skills and abilities.

Jobs are constantly changing and evolving, and employees of all ranks should show that they are open to growing and learning, either by experiencing new situations, by training, or even by listening and learning from criticism.

Even if your job has nothing to do with mathematics, arithmetic, geometry, algebra, calculus, and statistics, basic knowledge of these may become necessary at some stage.

Refreshing your knowledge of mathematics often is an essential part of keeping your competitive advantage in the job market. Statistics in particular may come in handy, as many a time you might need to produce some graphs and figures by analyzing quantitative data.

Being analytical, but also having strong research skills, differentiates one employee from the other. It demonstrates your determination, your ability to assess different scenarios, and your commitment to be 100% sure before giving an answer to your employer. It could mean the difference between a badly thought idea and something that may gain the company a huge profit. As much as you think a question/problem presented to you is a piece of cake, be very wary of giving a rushed answer. Take the time to analyze the situation, think of all possible scenarios, and if, possible, ask for some time to go and do some research to find out more.

Power BI for the Technology Industry

Convert Data into Insights with help of Power BI

Access your data easily on any device and get a 360-degree view of the health of your company. Power BI is a complete package of solutions tailored for the digital business world. From better team collaboration to interactive reports, data management, and predictive analysis – Power BI is the perfect intelligence tool for your business. The augmented growth of data in today's business landscape undoubtedly demands a reliable tool to manage heaps of data and reports efficiently. Power BI reconstructs the process instances step-by-step, even if they are from multiple sources, and helps you cut out unnecessary steps, get accurate results to interpret and act on, and reduce costs.

Power BI's advanced business intelligence and data modeling capabilities offer businesses the power of predictive analytics to compete in today's digital world. With features to generate historical data (past) and analyze current market trends and financial graphs, enterprises could easily prepare for the future. Power BI's

- Predictive analytics – use of historical data, patterns, and ML techniques to predict future outcomes and
- Prescriptive analytics – uses data and goes in-depth into the potential results of specific actions capabilities have significantly helped enterprises to arrive at better business decisions.

Connect to Data

Bring in your data from different Data Sources- be it on-premise or cloud data.

- Hundreds of Database Connectors
- Data Sources added on regular basis

Transform, Shape & Model Data

Get the connected data from various sources to work together without the need to flatten data.

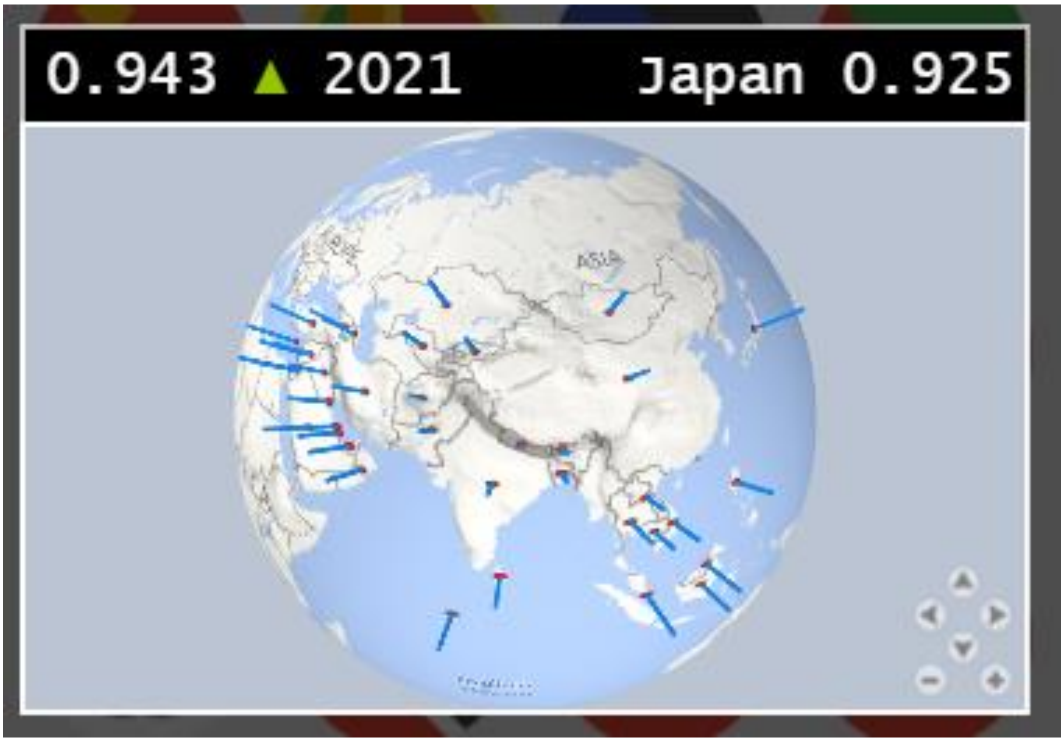
- Data Transformation
- Connects various Data Sources
- Data Model Optimization

Reports

Gain deeper insights into your data with rich interactive and meaningful reports.

- Data Visualization
- Quick Business insights
- Drill down functionality
- Real-time Goal Monitoring

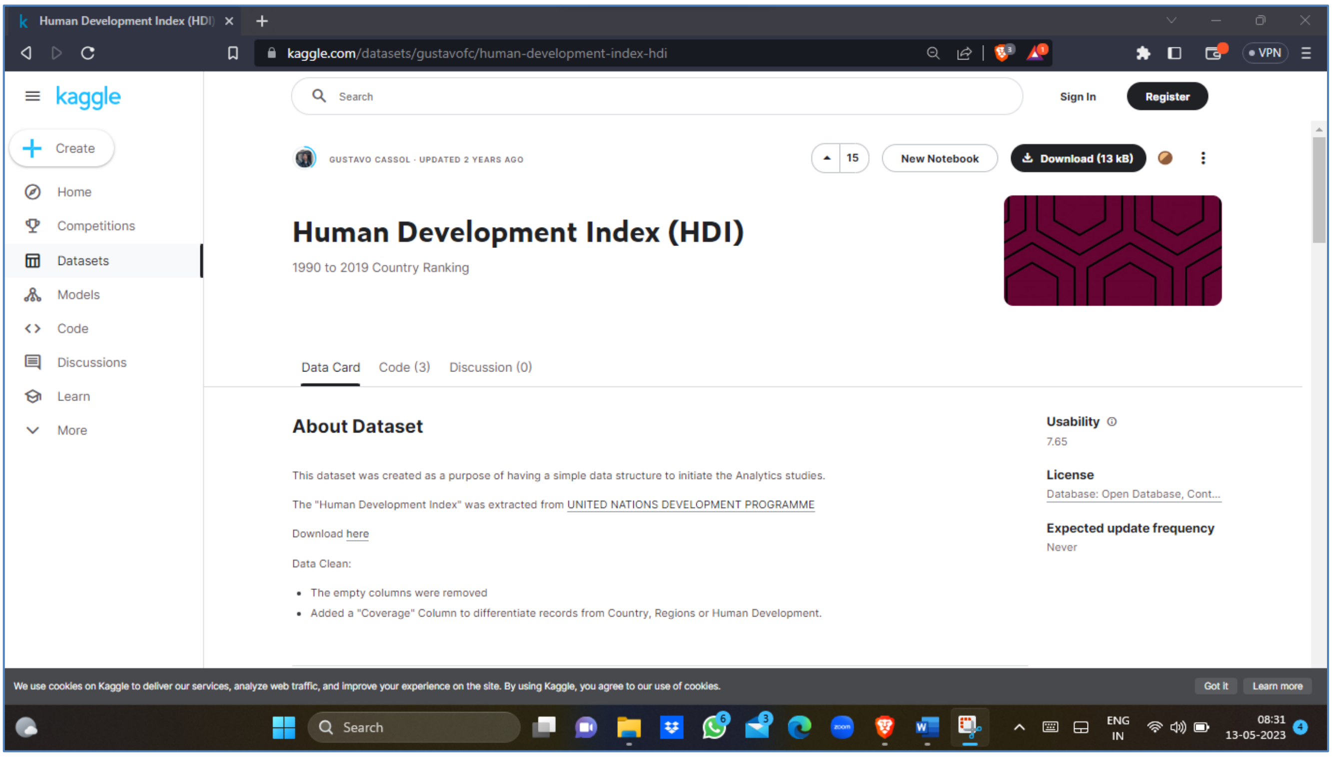
PROJECT
COUNTRIES RANKING ANALYSIS BUILT-IN POWER BI



About the Dataset
DATA SOURCE

Kaggle, a subsidiary of Google LLC, is an online community of data scientists and machine learning practitioners. Kaggle allows users to find and publish data sets, explore and build models in a web-based data-science environment, work with other data scientists and machine learning engineers, and enter competitions to solve data science challenges. Kaggle got its start in 2010 by offering machine learning and data science competitions as well as offering a public data and cloud-based business platform for data science and AI education. Most of the core staff were two Anthony Goldblum and Jeremy Howard.

DATASET – ***HUMAN DEVELOPENT INDEX DATA***



COLUMNS OF THE DATASET

- name
- alpha-2
- alpha-3
- country-code
- iso_3166-2
- region
- sub-region
- intermediate-region
- region-code
- sub-region code
- intermediate-region-code
- Importance
- Country
- Altitude
- Latitude
- Longitude
- Image_file
- Image url
- Growth_rate
- Rank
- Year
- Human Development Index
- Literacy rate
- Male Literacy rate
- Female Literacy rate
- Gender difference Percent

KEY POINTS

This dataset includes 50 thousand+ unique values which are categorized by Countries rankings based on HDI, GDP, Literacy rate . It also consists of various other attributes such as hdi rates, gdp rates, male & female literacy rates, etc. which further contribute to *Countries Ranking Analysis.*
In this data, we can see the hdi growth from year to year. This data is on the overall growth of Countries all over the . Each and every column is regarding the data and gives accurate information to the viewer. Through this, we can get good data and vi Visualization.

ORDERS

There are some columns in this table like;

- name
- alpha-2
- alpha-3
- country-code
- iso_3166-2
- region
- sub-region
- intermediate-region
- region-code
- sub-region code
- intermediate-region-code
- Importance
- Country
- Altitude
- Latitude
- Longitude
- Image_file
- Image url
- Growth_rate
- Rank
- Year
- Human Development Index
- Literacy rate
- Male Literacy rate
- Female Literacy rate
- Gender difference Percent

Like the above table, there are two other tables called the **people table** and the **return table**, in which there are columns like regions, country, returned or not etc and etc…

REPORT BUILDING

Visualization tool – Power BI

Microsoft Power BI is an interactive data visualization software product developed by Microsoft with a primary focus on business intelligence. It is part of the Microsoft Power Platform. Power BI is a collection of software services, apps, and connectors that work together to turn unrelated sources of data into coherent, visually immersive, and interactive insights. Data may be input by reading directly from a database, webpage, or structured files such as spreadsheets, CSV, XML, and JSON.
Power BI is a business and data analytics service that enables professionals to process, analyze, and visualize vast volumes of data. It helps extract insights, draw conclusions, and share results in the form of reports and dashboards across various departments. It provides an easy calculation or identifies future analysis or growth.

"Power BI desktop app is used to create reports, while Power BI Services (Software as a Service - Seas') is used to publish the reports, and Power BI mobile app is used to view the reports and dashboards."

Power BI dashboards are **single-page documents, often referred to as canvas, which illustrate a story through visualizations**. A well-designed dashboard consists only of the highlights of a story. This is due to the fact that it is limited to one page. For more information, readers can view related reports. Power BI is the only service that offers dashboards.

Single-page visualization with multiple charts and graphs to tell a story is called a Power BI dashboard. This one-page visualization in a dashboard is also known as a **Canvas**. The Power BI dashboard is a feature only available in Power BI Service. Since a Power BI dashboard is limited to one page, it only contains the highlights of a story. You cannot create a dashboard on Power BI Desktop.

Power BI is a Data Visualization and Business Intelligence tool that converts data from different data sources to interactive dashboards and BI reports. Power BI suite provides multiple software, connector, and services - Power BI desktop, Power BI service based on Saas, and mobile Power BI apps available for different platforms. This set of services is used by business users to consume data and build BI reports.

The Power BI desktop app is used to create reports, while Power BI Services (Software as a Service - SaaS) is used to publish the reports, and Power BI mobile app is used to view the reports and dashboards.

Microsoft Power BI Desktop is built for the analyst. It combines state-of-the-art interactive visualizations, with industry-leading data query and modeling built-in. Create and publish your reports to Power BI. Power BI Desktop helps you empower others with timely critical insights, anytime, anywhere.

With Power BI Desktop, you can:

- **Get data**
    - The Power BI Desktop makes discovering data easy. You can import data from a wide variety of data sources. After you connect to a data source, you can shape the data to match your analysis and reporting needs.
- **Create relationships and enrich your data model with new measures and data formats**
    - When you import two or more tables, oftentimes you'll need to create relationships between those tables. The Power BI Desktop includes the Manage Relationships dialog and the Relationships view, where you can use Autodetect to let the Power BI Desktop find and create any relationships, or you can create them yourself. You can also very easily create your own measures and calculations or customize data formats and categories to enrich your data for additional insights.
- **Create reports**
    - The Power BI Desktop includes the Report View. Select the fields you want, add filters, choose from dozens of visualizations, format your reports with custom colors, gradients and several other options. The Report View gives you the same great report and visualizations tools just like when creating a report on PowerBI.com.
- **Save your reports**
    - With the Power BI Desktop, you can save your work as a Power BI Desktop file. Power BI Desktop files have a .pbix extension.
- **Upload or Publish your reports**
    - You can upload the reports you created and saved in the Desktop to your Power BI site. You can also publish them to Power BI right from Power BI Desktop.

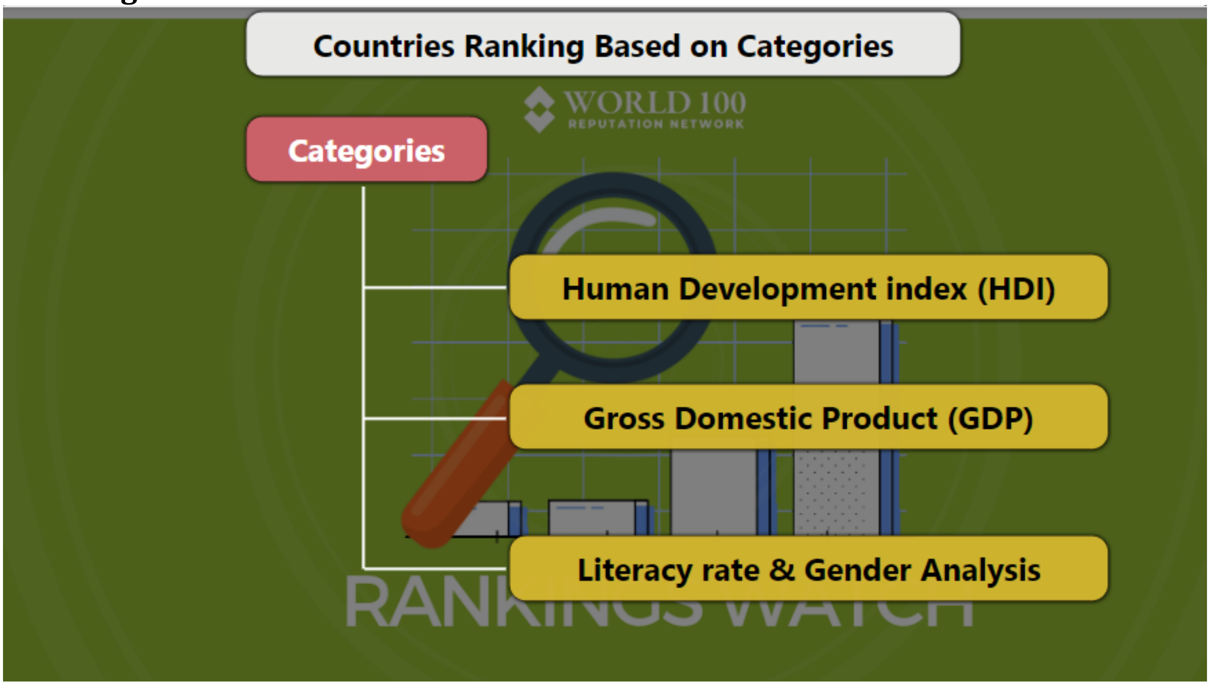
VISUALIZATION

Topic: Countries Ranking Based on Categories

1) What is my Data??

I have taken data on Human Development Index (HDI), Gross Domestic Product (GDP), Literacy rate & Gender Analysis Rankings

Main Page



2) Why I taken this Data??

To know the countries Rankings on basis of HDI, GDP and Literacy rate (Male, Female) in Detailed view

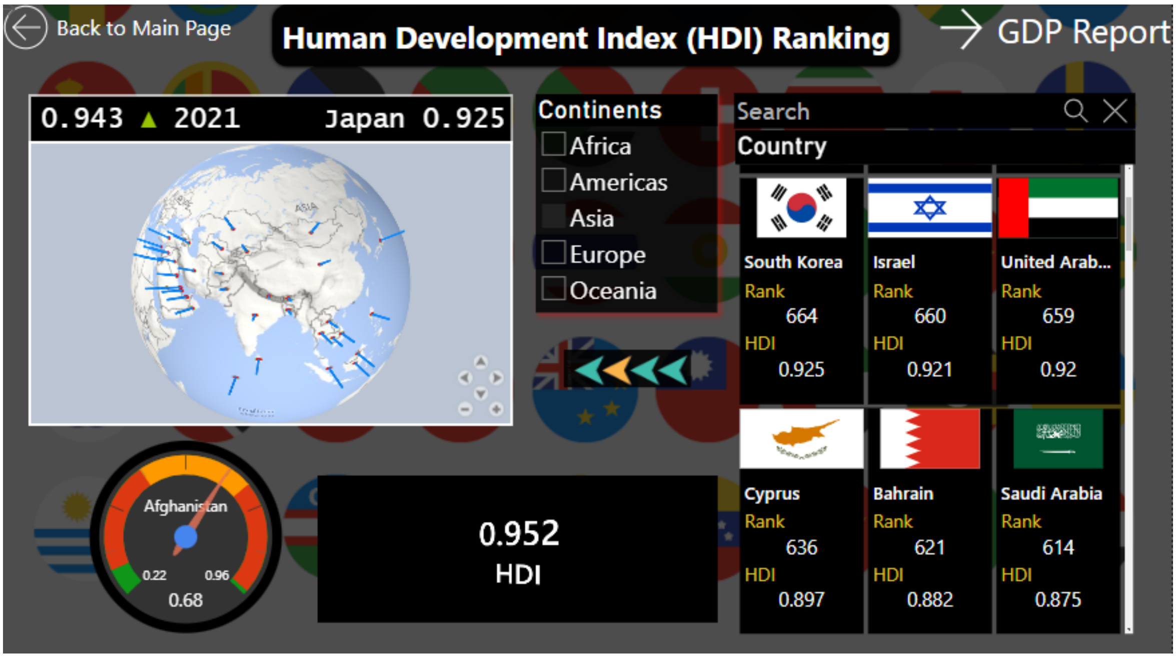
Tools used:

Tools used in this Main Page are given below,

1. **Canvas Background:** This setting is used to set the background image for the page which improves the data appearance.

2. **Shapes:** This shape tool is used to set the titles and give actions like page navigations, etc.,

HDI PAGE



3) What is the use of this HDI Page??

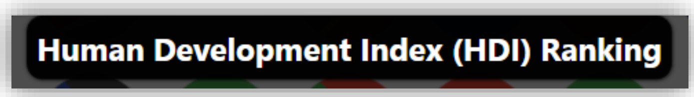
We can see Human Development Index (HDI) rate and Rankings in full detailed visualisation here. By this page we can custom search the country name and this displays location, continent flag, tachometer reading and Rank of that Country

Tools used:

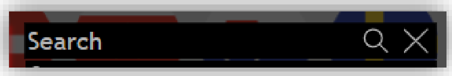
Tools used in this Main Page are given below,

1. **Canvas Background:** This setting is used to set the background image for the page which improves the data appearance.

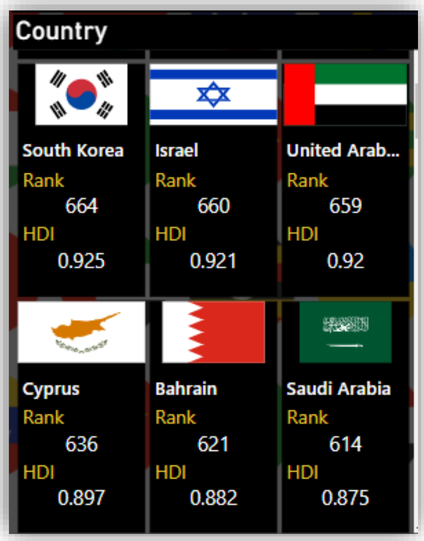
2. **Shapes:** This shape tool is used to set the titles and give actions like page navigations, etc.,



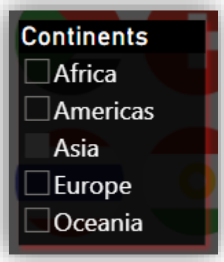
3. **Text Search Slicer:** This tool is used to search a specific option from the list in the data



4. **Multi Info Cards:** This tool is used to preview the multiple information with picture support, So that you can provide picture url and it will provide picture for that.

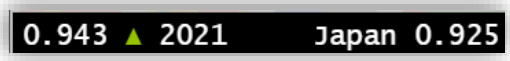


5. **Slicer:** Slicer tool is used to categorize the data into different forms.



6. **Rotating Card:** This tool is used to visualise different column information in one page.

7. **Scroller:** This scroller tool is used to visualise data in scrolling form.



8. **Globe map:** This globe map tool is used to view certain location in globe map.

9. **Dial Gauge:** This tool is used to understand easily the position of the specific category we pick from data.

GDP PAGE

4) What is the use of this GDP Page??

We can see Gross Domestic Product (GDP) rate and Rankings in full detailed visualisation here. By this page we can custom search the country name and this displays location, continent flag, GDP Reading, and Rank of that Country

Tools used:

Tools used in this Main Page are given below,

1. **Canvas Background:** This setting is used to set the background image for the page which improves the data appearance

2. **Shapes:** This shape tool is used to set the titles and give actions like page navigations, etc.,

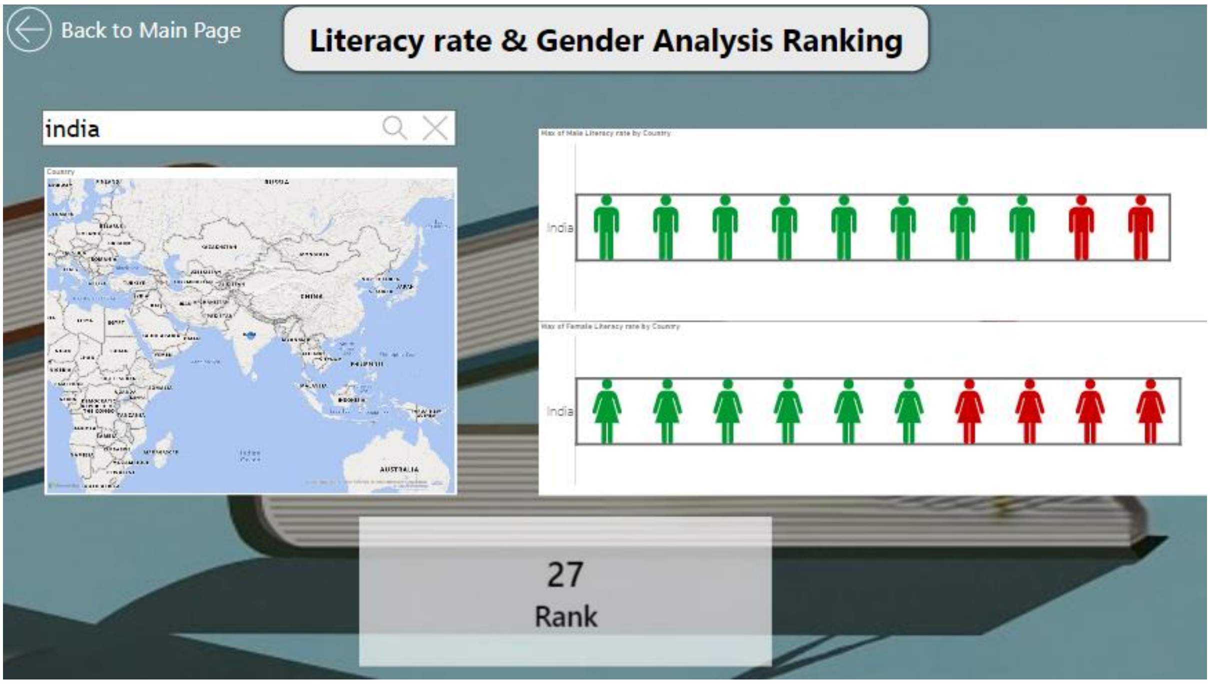
3. **Rotating Card:** This tool is used to visualise different column information in one page.

4. **Multi Info Cards:** This tool is used to preview the multiple information with picture support, So that you can provide picture url and it will provide picture for that.

5. **Waterdrop Map:** This tool is used to view specific location in map

6. **Text Search Slicer:** This tool is used to search a specific option from the list in the data

LITERACY RATE PAGE



5) What is the use of this Literacy Rate Page??

We can see Literacy rate and Gender Analysis Rankings in full detailed visualisation here. By this page we can custom search the country name and this displays location, Male and Female visual Difference and Rank of that Country

Tools used:

Tools used in this Main Page are given below,

1. **Canvas Background:** This setting is used to set the background image for the page which improves the data appearance

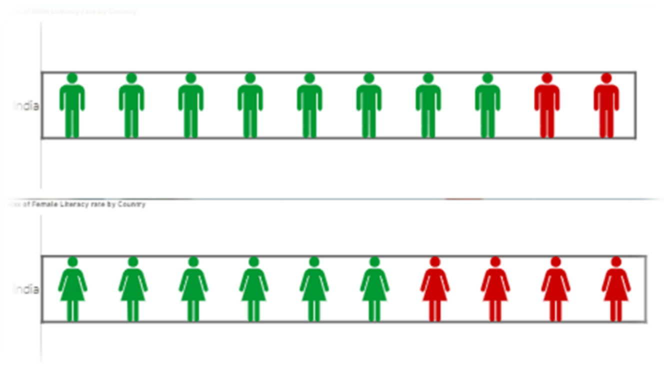
2. **Shapes:** This shape tool is used to set the titles and give actions like page navigations, etc.,

3. **Map:** This tool is used to view specific location in map

4. **Text Search Slicer:** This tool is used to search a specific option from the list in the data

5. **Rotating Card:** This tool is used to visualise different column information in one page.

6. **Infographic designer:** This tool is used to visualise the percentage of the data in image form.



CONCLUSION

By this project we can get the Rankings of any countries in the world based on different categories like Human Development Index(HDI), Gross domestic Product (GDP), Literacy rate in gender analysis in male and female

-

REFERENCES

https://www.kaggle.com/
https://powerbi.microsoft.com/en-au/
https://en.wikipedia.org/wiki/Good_Country_Index
https://www.kaggle.com/datasets
https://www.kaggle.com/datasets/gustavofc/human-development-index-hdi

THANK YOU.

EVALUATION

Student Self Evaluation of the Short-Term Internship

Student Name: **Registration No:**

Term of Internship: **From:** **To :**

Date of Evaluation:

Organization Name & Address:

Please rate your performance in the following areas:

Rating Scale: **Letter grade of CGPA calculation to be provided**

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	**OVERALL PERFORMANCE**	**1**	**2**	**3**	**4**	**5**

Date: **Signature of the Studen**

Evaluation by the Supervisor of the Intern Organization

Student Name:		**Registration No:**
Term of Internship:	**From:**	**To :**
Date of Evaluation:		
Organization Name & Address:		
Name & Address of the Supervisor with Mobile Number		

Please rate the student's performance in the following areas:

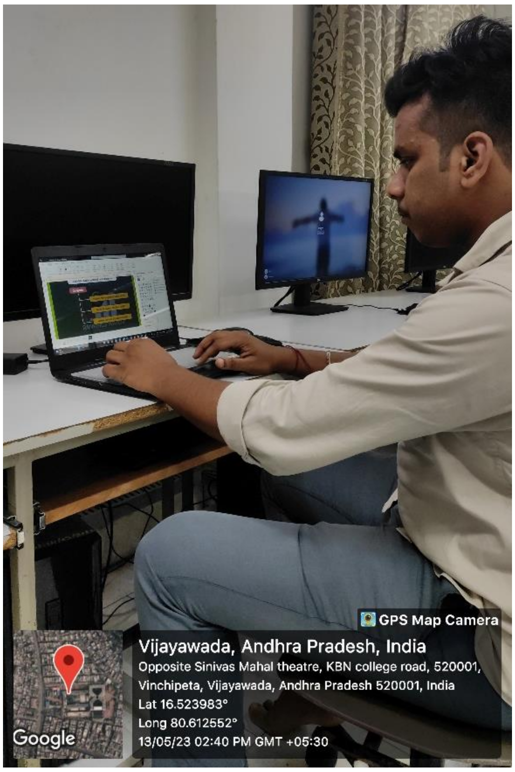
Please note that your evaluation shall be done independent of the Student's self-evaluation

Rating Scale: 1 is lowest and 5 is highest rank

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	**OVERALL PERFORMANCE**	**1**	**2**	**3**	**4**	**5**

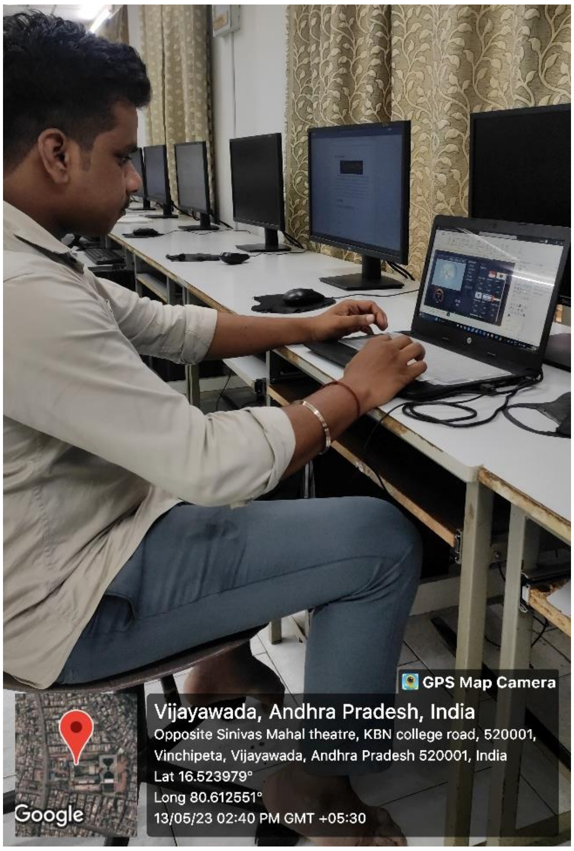
Date: **Signature of the Supervisor**

PHOTOS & VIDEO LINKS

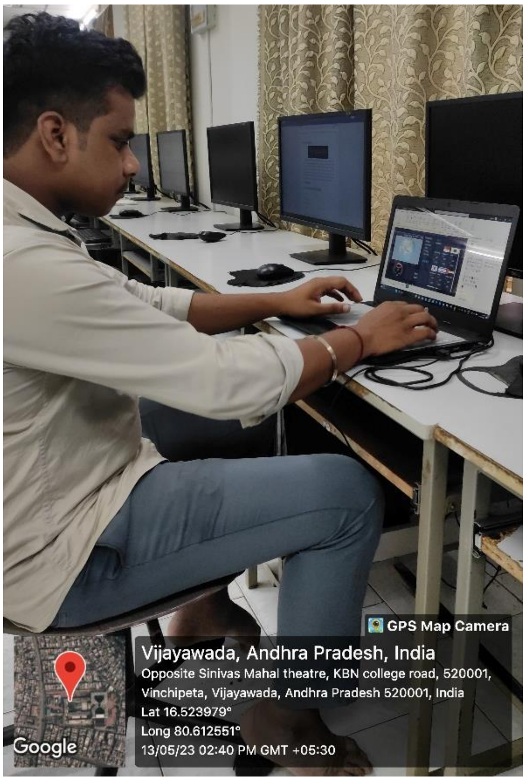

GPS Map Camera
Google
Vijayawada, Andhra Pradesh, India
Opposite Sinivas Mahal theatre, KBN college road, 520001,
Vinchipeta, Vijayawada, Andhra Pradesh 520001, India
Lat 16.523983°
Long 80.612552°
13/05/23 02:40 PM GMT +05:30


GPS Map Camera
Google
Vijayawada, Andhra Pradesh, India
Opposite Sinivas Mahal theatre, KBN college road, 520001,
Vinchipeta, Vijayawada, Andhra Pradesh 520001, India
Lat 16.523995°
Long 80.612529°
06/05/23 03:28 PM GMT +05:30


GPS Map Camera
Google
Vijayawada, Andhra Pradesh, India
Opposite Sinivas Mahal theatre, KBN college road, 520001,
Vinchipeta, Vijayawada, Andhra Pradesh 520001, India
Lat 16.523966°
Long 80.612519°
06/05/23 03:25 PM GMT +05:30


GPS Map Camera
Google
Vijayawada, Andhra Pradesh, India
Opposite Sinivas Mahal theatre, KBN college road, 520001,
Vinchipeta, Vijayawada, Andhra Pradesh 520001, India
Lat 16.523979°
Long 80.612551°
13/05/23 02:40 PM GMT +05:30


GPS Map Camera
Vijayawada, Andhra Pradesh, India
Opposite Sinivas Mahal theatre, KBN college road, 520001,
Vinchipeta, Vijayawada, Andhra Pradesh 520001, India
Lat 16.524004°
Long 80.612519°
06/05/23 03:26 PM GMT +05:30
Google



Video Link : https://drive.google.com/drive/folders/1khRUNt3eaEUA_3xq8PiJHJcnyZXDaqmn?usp=sharing

MARKS STATEMENT
(To be used by the Examiners)

INTERNAL ASSESSMENT STATEMENT

Name Of the Student:
Programme of Study:
Year of Study:
Group:
Register No/H.T. No:
Name of the College:
University:

Sl.No	*Evaluation Criterion*	*Maximum Marks*	*Marks Awarded*
1.	Activity Log	10	
2.	Internship Evaluation	30	
3.	Oral Presentation	10	
	GRAND TOTAL	50	

Date: Signature of the Faculty Guide

EXTERNAL ASSESSMENT STATEMENT

Name Of the Student:
Programme of Study:
Year of Study:
Group:
Register No/H.T. No:
Name of the College:
University:

Sl.No	*Evaluation Criterion*	*Maximum Marks*	*Marks Awarded*
1.	Internship Evaluation	80	
2.	For the grading giving by the Supervisor of the Intern Organization	20	
3.	Viva-Voce	50	
	TOTAL	150	
GRAND TOTAL (EXT. 50 M + INT. 100M)		**200**	

Signature of the Faculty Guide

Signature of the Internal Expert

Signature of the External Expert

Signature of the Principal with Seal